

NCERT Exemplar Solutions For Grade 12

Mathematics

Chapter 4. - Determinants

Solved Examples

Short Answer (S. A)

Example 1: If $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & 5 \\ 8 & 3 \end{vmatrix}$, then find x .

Ans: here, we have $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & 5 \\ 8 & 3 \end{vmatrix}$

$$\Rightarrow 2x^2 - 40 = 18 - 40$$

$$\Rightarrow 2x^2 = 18$$

$$\Rightarrow x^2 = \frac{18}{2}$$

$$\Rightarrow x^2 = 9$$

$$\Rightarrow x = \sqrt{9}$$

$$\Rightarrow x = \pm 3$$

Hence, value of x is ± 3 .

Example 2: If $\Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$, $\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ yz & zx & xy \\ x & y & z \end{vmatrix}$, then prove that

$$\Delta + \Delta_1 = 0.$$

Ans: here, we have $\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ yz & zx & xy \\ x & y & z \end{vmatrix}$

Now, interchanging rows and columns, we get

$$\Rightarrow \Delta_1 = \begin{vmatrix} 1 & yz & x \\ 1 & zx & y \\ 1 & xy & z \end{vmatrix}$$

$$\Rightarrow \Delta_1 = \frac{1}{xyz} \begin{vmatrix} x & xyz & x^2 \\ y & xyz & y^2 \\ z & xyz & z^2 \end{vmatrix}$$

Taking common xyz from C_2 , we get

$$\Rightarrow \Delta_1 = \frac{1}{xyz} \times xyz \begin{vmatrix} x & 1 & x^2 \\ y & 1 & y^2 \\ z & 1 & z^2 \end{vmatrix}$$

Now, interchanging C_1 and C_2 , we get

$$\Rightarrow \Delta_1 = (-1) \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$$\Rightarrow \Delta_1 = -\Delta$$

$$\Rightarrow \Delta + \Delta_1 = 0$$

Hence proved.

Example 3: Without expanding, Show that $\Delta = \begin{vmatrix} \operatorname{cosec}^2 \theta & \cot^2 \theta & 1 \\ \cot^2 \theta & \operatorname{cosec}^2 \theta & -1 \\ 42 & 40 & 2 \end{vmatrix} = 0$

Ans: here, we have $\Delta = \begin{vmatrix} \operatorname{cosec}^2\theta & \cot^2\theta & 1 \\ \cot^2\theta & \operatorname{cosec}^2\theta & -1 \\ 42 & 40 & 2 \end{vmatrix}$

Applying $C_1 \rightarrow C_1 - C_2$

$$\Rightarrow \Delta = \begin{vmatrix} \operatorname{cosec}^2\theta - \cot^2\theta & \cot^2\theta & 1 \\ \cot^2\theta - \operatorname{cosec}^2\theta & \operatorname{cosec}^2\theta & -1 \\ 2 & 40 & 2 \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} \operatorname{cosec}^2\theta - \cot^2\theta & \cot^2\theta & 1 \\ -(\operatorname{cosec}^2\theta - \cot^2\theta) & \operatorname{cosec}^2\theta & -1 \\ 2 & 40 & 2 \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} 1 & \cot^2\theta & 1 \\ -1 & \operatorname{cosec}^2\theta & -1 \\ 2 & 40 & 2 \end{vmatrix}$$

$$\Rightarrow \Delta = 0 \quad [\text{when two columns are same, then value of determinants is } 0]$$

Hence proved.

Example 4: Show that $\Delta = \begin{vmatrix} x & p & q \\ p & x & q \\ q & q & x \end{vmatrix} = (x - p)(x^2 + px - 2q^2)$.

Ans: here, we have $\Delta = \begin{vmatrix} x & p & q \\ p & x & q \\ q & q & x \end{vmatrix}$

Applying $C_1 \rightarrow C_1 - C_2$

$$\Rightarrow \Delta = \begin{vmatrix} x-p & p & q \\ -(x-p) & x & q \\ 0 & q & x \end{vmatrix}$$

Taking common $(x - p)$ from C_1

$$\Rightarrow \Delta = (x-p) \begin{vmatrix} 1 & p & q \\ -1 & x & q \\ 0 & q & x \end{vmatrix}$$

Now, expanding along C_1

$$\Rightarrow \Delta = (x-p) [1(x^2 - q^2) + 1(px - q^2) + 0]$$

$$\Rightarrow \Delta = (x-p)(x^2 + px - 2q^2)$$

Hence proved.

Example 5: If $\Delta = \begin{vmatrix} 0 & b-a & c-a \\ a-b & 0 & c-b \\ a-c & b-c & 0 \end{vmatrix}$, then show that Δ is equal to zero.

Ans: here, we have $\Delta = \begin{vmatrix} 0 & b-a & c-a \\ a-b & 0 & c-b \\ a-c & b-c & 0 \end{vmatrix}$

Taking “-1” as common from R_1 , R_2 and R_3 , we get

$$\Rightarrow \Delta = (-1)^3 \begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$$

Now, interchanging rows and columns, we get

$$\Rightarrow \Delta = (-1) \begin{vmatrix} 0 & b-a & c-a \\ a-b & 0 & c-b \\ a-c & b-c & 0 \end{vmatrix}$$

$$\Rightarrow \Delta = -\Delta$$

$$\Rightarrow 2\Delta = 0$$

$$\Rightarrow \Delta = 0$$

Hence proved.

Example 6: Prove that $(A^{-1})' = (A')^{-1}$, where A is an invertible matrix.

Ans: here, matrix A is an invertible matrix, so it is non – singular i.e., $|A| \neq 0$.

And we know that, $|A| = |A'|$. So, $|A'| \neq 0$.

Also, we have $AA^{-1} = A^{-1}A = I$

Taking transpose on both sides, we get

$$\Rightarrow (AA^{-1})' = (A^{-1}A)' = I'$$

$$\Rightarrow (A^{-1})' A' = A' (A^{-1})' = I$$

Thus, $(A^{-1})'$ is inverse of A' ,

Therefore, $(A^{-1})' = (A')^{-1}$

Hence proved.

Long Answer (L.A.)

Example 7: If $x = -4$ is a root of $\Delta = \begin{vmatrix} x & 2 & 3 \\ 1 & x & 1 \\ 3 & 2 & x \end{vmatrix} = 0$, then find the other two roots.

Ans: here, we have $\begin{vmatrix} x & 2 & 3 \\ 1 & x & 1 \\ 3 & 2 & x \end{vmatrix} = 0$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$

$$\Rightarrow \begin{vmatrix} x+4 & x+4 & x+4 \\ 1 & x & 1 \\ 3 & 2 & x \end{vmatrix} = 0$$

Taking $(x+4)$ common from R_1

$$\Rightarrow (x+4) \begin{vmatrix} 1 & 1 & 1 \\ 1 & x & 1 \\ 3 & 2 & x \end{vmatrix} = 0$$

Applying $C_1 \rightarrow C_1 - C_3$ and $C_2 \rightarrow C_2 - C_3$

$$\Rightarrow (x+4) \begin{vmatrix} 0 & 0 & 1 \\ 0 & x-1 & 1 \\ 3-x & 2-x & x \end{vmatrix} = 0$$

$$\Rightarrow (x+4)[0-(3-x)(x-1)] = 0$$

$$\Rightarrow (x+4)(3-x)(x-1) = 0$$

$$\Rightarrow x = -4, 3, 1$$

Hence, other two roots are 1 and 3.

Example 8: In a triangle ABC, if $\begin{vmatrix} 1 & 1 & 1 \\ 1 + \sin A & 1 + \sin B & 1 + \sin C \\ \sin A + \sin^2 A & \sin B + \sin^2 B & \sin C + \sin^2 C \end{vmatrix} = 0$,

then prove that ΔABC is an isosceles triangle.

Ans: here, we have $\begin{vmatrix} 1 & 1 & 1 \\ 1 + \sin A & 1 + \sin B & 1 + \sin C \\ \sin A + \sin^2 A & \sin B + \sin^2 B & \sin C + \sin^2 C \end{vmatrix} = 0$

$$\Rightarrow \text{Applying } C_1 \rightarrow C_1 - C_3 \text{ and } C_2 \rightarrow C_2 - C_3$$

$$\Rightarrow \begin{vmatrix} 0 & 0 & 1 \\ \sin A - \sin B & \sin B - \sin C & 1 + \sin C \\ \sin A - \sin B + \sin^2 A - \sin^2 B & \sin B - \sin C + \sin^2 B - \sin^2 C & \sin C + \sin^2 C \end{vmatrix} = 0$$

Taking $(\sin A - \sin B)$ and $(\sin B - \sin C)$ from C_1 and C_2 respectively

$$\Rightarrow (\sin A - \sin B)(\sin B - \sin C) \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 + \sin C \\ 1 + \sin A + \sin B & 1 + \sin B + \sin C & \sin C + \sin^2 C \end{vmatrix} = 0$$

Now, expanding along R_1 , we get

$$\Rightarrow (\sin A - \sin B)(\sin B - \sin C)(1 + \sin B + \sin C - 1 - \sin A - \sin B) = 0$$

$$\Rightarrow (\sin A - \sin B)(\sin B - \sin C)(\sin C - \sin A) = 0$$

$$\Rightarrow (\sin A - \sin B) = 0 \text{ or } (\sin B - \sin C) = 0 \text{ or } (\sin C - \sin A) = 0$$

$$\Rightarrow \sin A = \sin B \text{ or } \sin B = \sin C \text{ or } \sin C = \sin A$$

$$\Rightarrow A = B \text{ or } B = C \text{ or } C = A$$

Hence, triangle ABC is an isosceles triangle.

Hence proved.

Example 9: Show that if the determinant $\Delta = \begin{vmatrix} 3 & -2 & \sin 3\theta \\ -7 & 8 & \cos 2\theta \\ -11 & 14 & 2 \end{vmatrix} = 0$, then $\sin \theta = 0$

or $\frac{1}{2}$

Ans: here, we have $\begin{vmatrix} 3 & -2 & \sin 3\theta \\ -7 & 8 & \cos 2\theta \\ -11 & 14 & 2 \end{vmatrix} = 0$

Applying $R_2 \rightarrow R_2 + 4R_1$ and $R_3 \rightarrow R_3 + 7R_1$, we get

$$\Rightarrow \begin{vmatrix} 3 & -2 & \sin 3\theta \\ 5 & 0 & \cos 2\theta + 4\sin 3\theta \\ 10 & 0 & 2 + 7\sin 3\theta \end{vmatrix} = 0$$

Now, expanding along C_2 , we get

$$\Rightarrow 2 [5(2 + 7\sin 3\theta) - 10(\cos 2\theta + 4\sin 3\theta)] = 0$$

$$\Rightarrow [10 + 35\sin 3\theta - 10\cos 2\theta - 40\sin 3\theta] = 0$$

$$\Rightarrow 10 - 5\sin 3\theta - 10\cos 2\theta = 0$$

$$\Rightarrow 2 - \sin 3\theta - 2\cos 2\theta = 0$$

$$\Rightarrow 2 - (3\sin\theta - 4\sin^3\theta) - 2(1 - 2\sin^2\theta) = 0$$

$$\Rightarrow 2 - 3\sin\theta + 4\sin^3\theta - 2 + 4\sin^2\theta = 0$$

$$\Rightarrow -3\sin\theta + 4\sin^3\theta + 4\sin^2\theta = 0$$

$$\Rightarrow \sin\theta(-3 + 4\sin^2\theta + 4\sin\theta) = 0$$

$$\Rightarrow \sin\theta(4\sin^2\theta + 4\sin\theta - 3) = 0$$

$$\Rightarrow \sin\theta(4\sin^2\theta + 6\sin\theta - 2\sin\theta - 3) = 0$$

$$\Rightarrow \sin\theta[2\sin\theta(2\sin\theta + 3) - 1(2\sin\theta + 3)] = 0$$

$$\Rightarrow \sin\theta(2\sin\theta + 3)(2\sin\theta - 1) = 0$$

$$\Rightarrow \sin\theta = 0 \text{ or } (2\sin\theta + 3) = 0 \text{ or } (2\sin\theta - 1) = 0$$

$$\Rightarrow \sin\theta = 0 \text{ or } \sin\theta = -\frac{3}{2} \text{ or } \sin\theta = \frac{1}{2}$$

As we know that $\sin\theta \in [-1, 1]$

Thus, $\sin\theta = 0$ or $\sin\theta = \frac{1}{2}$

Hence proved.

Objective Type Questions

Choose the correct answer from the given four options in each of the Example 10 and 11.

Example 10: Let $\Delta = \begin{vmatrix} Ax & x^2 & 1 \\ By & y^2 & 1 \\ Cz & z^2 & 1 \end{vmatrix}$ and $\Delta_1 = \begin{vmatrix} A & B & C \\ x & y & z \\ zy & zx & xy \end{vmatrix}$, then

- (A) $\Delta_1 = \Delta$
- (B) $\Delta \neq \Delta_1$
- (C) $\Delta - \Delta_1 = 0$
- (D) None of these

Ans: here, we have $\Delta = \begin{vmatrix} Ax & x^2 & 1 \\ By & y^2 & 1 \\ Cz & z^2 & 1 \end{vmatrix}$ and $\Delta_1 = \begin{vmatrix} A & B & C \\ x & y & z \\ zy & zx & xy \end{vmatrix}$

$$\Rightarrow \Delta_1 = \begin{vmatrix} A & B & C \\ x & y & z \\ zy & zx & xy \end{vmatrix}$$

Now, interchanging rows and columns, we get

$$\Rightarrow \Delta_1 = \begin{vmatrix} A & x & zy \\ B & y & zx \\ C & z & xy \end{vmatrix}$$

$$\Rightarrow \Delta_1 = \frac{1}{xyz} \begin{vmatrix} Ax & x^2 & xyz \\ By & y^2 & xyz \\ Cz & z^2 & xyz \end{vmatrix}$$

Taking xyz common from C_3 , we get

$$\Rightarrow \Delta_1 = \frac{1}{xyz} \times xyz \begin{vmatrix} Ax & x^2 & 1 \\ By & y^2 & 1 \\ Cz & z^2 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta_1 = \Delta$$

$$\Rightarrow \Delta - \Delta_1 = 0$$

Hence, option (C) is the correct answer.

Example 11: If $x, y \in R$, then the determinant

$$\Delta = \begin{vmatrix} \cos x & -\sin x & 1 \\ \sin x & \cos x & 1 \\ \cos(x+y) & -\sin(x+y) & 0 \end{vmatrix} \text{ lies in the interval}$$

(A) $[-\sqrt{2}, \sqrt{2}]$

(B) $[-1, 1]$

(C) $[-\sqrt{2}, 1]$

(D) $[-1, -\sqrt{2}]$

Ans: here, we have $\Delta = \begin{vmatrix} \cos x & -\sin x & 1 \\ \sin x & \cos x & 1 \\ \cos(x+y) & -\sin(x+y) & 0 \end{vmatrix}$

Applying $R_3 \rightarrow R_3 - R_1 \cos y + R_2 \sin y$

$$\Rightarrow \Delta = \begin{vmatrix} \cos x & -\sin x & 1 \\ \sin x & \cos x & 1 \\ 0 & 0 & \sin y - \cos y \end{vmatrix}$$

Now, expanding along R_3 , we get

$$\Rightarrow \Delta = (\sin y - \cos y)(\cos^2 x + \sin^2 x)$$

$$\Rightarrow \Delta = (\sin y - \cos y)$$

$$\Rightarrow \Delta = \sqrt{2} \left(\frac{1}{\sqrt{2}} \sin y - \frac{1}{\sqrt{2}} \cos y \right)$$

$$\Rightarrow \Delta = \sqrt{2} \left(\cos \frac{\pi}{4} \sin y - \sin \frac{\pi}{4} \cos y \right)$$

$$\Rightarrow \Delta = \sqrt{2} \sin \left(y - \frac{\pi}{4} \right)$$

As we know that, $-1 \leq \sin \left(y - \frac{\pi}{4} \right) \leq 1$

$$\Rightarrow -\sqrt{2} \leq \sqrt{2} \sin \left(y - \frac{\pi}{4} \right) \leq \sqrt{2}$$

$$\Rightarrow -\sqrt{2} \leq \Delta \leq \sqrt{2}$$

Hence, option (A) is the correct answer.

Fill in the blank in each of the examples 12 to 14.

Example 12: If A, B, C are the angles of a triangles, then $\Delta = \begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix} =$

.....

Ans: here, we have $\Delta = \begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$

Applying $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$, we get

$$\Rightarrow \Delta = \begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B - \sin^2 A & \cot B - \cot A & 0 \\ \sin^2 C - \sin^2 A & \cot C - \cot A & 0 \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin(B+A)\sin(B-A) & \frac{\cos B}{\sin B} - \frac{\cos A}{\sin A} & 0 \\ \sin(C+A)\sin(C-A) & \frac{\cos C}{\sin C} - \frac{\cos A}{\sin A} & 0 \end{vmatrix}$$

Here, $A + B + C = \pi$

$$\Rightarrow \Delta = \begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin(\pi-C)\sin(B-A) & \frac{\cos B \cdot \sin A - \cos A \cdot \sin B}{\sin B \cdot \sin A} & 0 \\ \sin(\pi-B)\sin(C-A) & \frac{\cos C \cdot \sin A - \cos A \cdot \sin C}{\sin C \cdot \sin A} & 0 \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} \sin^2 A & \frac{\cos A}{\sin A} & 1 \\ \sin C \cdot \sin(B-A) & \frac{\sin(A-B)}{\sin B \cdot \sin A} & 0 \\ \sin B \cdot \sin(C-A) & \frac{\sin(A-C)}{\sin C \cdot \sin A} & 0 \end{vmatrix}$$

Taking $\sin(B-A)$ and $\sin(C-A)$ common from R_2 and R_3 respectively.

$$\Rightarrow \Delta = \sin(B-A)\sin(C-A) \begin{vmatrix} \sin^2 A & \frac{\cos A}{\sin A} & 1 \\ \sin C & \frac{-1}{\sin B \cdot \sin A} & 0 \\ \sin B & \frac{-1}{\sin C \cdot \sin A} & 0 \end{vmatrix}$$

Taking $\frac{1}{\sin A}$ common from C_2 , we get

$$\Rightarrow \Delta = \sin(B-A)\sin(C-A) \begin{vmatrix} \sin^2 A & \cos A & 1 \\ \sin C & \frac{-1}{\sin B} & 0 \\ \sin B & \frac{-1}{\sin C} & 0 \end{vmatrix}$$

Now, expanding along R_1 , we get

$$\Rightarrow \Delta = \sin(B-A)\sin(C-A) (-1+1)$$

$$\Rightarrow \Delta = \sin(B-A)\sin(C-A) \times 0$$

$$\Rightarrow \Delta = 0$$

$$\text{Thus, } \Delta = \begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix} = 0.$$

Example 13: The determinant $\Delta = \begin{vmatrix} \sqrt{23} + \sqrt{3} & \sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{46} & 5 & \sqrt{10} \\ 3 + \sqrt{115} & \sqrt{15} & 5 \end{vmatrix}$ is equal to

Ans: here, we have $\Delta = \begin{vmatrix} \sqrt{23} + \sqrt{3} & \sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{46} & 5 & \sqrt{10} \\ 3 + \sqrt{115} & \sqrt{15} & 5 \end{vmatrix}$

Taking $\sqrt{5}$ common from C_2 and C_3 , we get

$$\Rightarrow \Delta = \sqrt{5} \times \sqrt{5} \begin{vmatrix} \sqrt{23} + \sqrt{3} & 1 & 1 \\ \sqrt{15} + \sqrt{46} & \sqrt{5} & \sqrt{2} \\ 3 + \sqrt{115} & \sqrt{3} & \sqrt{5} \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 - \sqrt{3}C_2 - \sqrt{23}C_3$

$$\Rightarrow \Delta = 5 \begin{vmatrix} 0 & 1 & 1 \\ 0 & \sqrt{5} & \sqrt{2} \\ 0 & \sqrt{3} & \sqrt{5} \end{vmatrix}$$

$\Rightarrow \Delta = 0$ [all the elements of a row is 0]

Thus, The determinant $\Delta = \begin{vmatrix} \sqrt{23} + \sqrt{3} & \sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{46} & 5 & \sqrt{10} \\ 3 + \sqrt{115} & \sqrt{15} & 5 \end{vmatrix}$ is equal to 0.

Example 14: The value of the determinant

$$\Delta = \begin{vmatrix} \sin^2 23^\circ & \sin^2 67^\circ & \cos 180^\circ \\ -\sin^2 67^\circ & -\sin^2 23^\circ & \cos^2 180^\circ \\ \cos 180^\circ & \sin^2 23^\circ & \sin^2 67^\circ \end{vmatrix} = \dots$$

Ans: here, we have $\Delta = \begin{vmatrix} \sin^2 23^\circ & \sin^2 67^\circ & \cos 180^\circ \\ -\sin^2 67^\circ & -\sin^2 23^\circ & \cos^2 180^\circ \\ \cos 180^\circ & \sin^2 23^\circ & \sin^2 67^\circ \end{vmatrix}$

$$\Rightarrow \Delta = \begin{vmatrix} \sin^2 23^\circ & \sin^2 (90-23)^\circ & -1 \\ -\sin^2 67^\circ & -\sin^2 (90-67)^\circ & 1 \\ -1 & \sin^2 23^\circ & \sin^2 67^\circ \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} \sin^2 23^\circ & \cos^2 23^\circ & -1 \\ -\sin^2 67^\circ & -\cos^2 67^\circ & 1 \\ -1 & \sin^2 23^\circ & \sin^2 67^\circ \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$

$$\Rightarrow \Delta = \begin{vmatrix} \sin^2 23^\circ + \cos^2 23^\circ - 1 & \cos^2 23^\circ & -1 \\ -(\sin^2 67^\circ + \cos^2 67^\circ) + 1 & -\cos^2 67^\circ & 1 \\ -1 + \sin^2 23^\circ + \sin^2 67^\circ & \sin^2 23^\circ & \sin^2 67^\circ \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} 1-1 & \cos^2 23^\circ & -1 \\ -1+1 & -\cos^2 67^\circ & 1 \\ -1+1 & \sin^2 23^\circ & \sin^2 67^\circ \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} 0 & \cos^2 23^\circ & -1 \\ 0 & -\cos^2 67^\circ & 1 \\ 0 & \sin^2 23^\circ & \sin^2 67^\circ \end{vmatrix}$$

$\Rightarrow \Delta = 0$ [all the elements of a column is 0]

$$\text{Hence, } \Delta = \begin{vmatrix} \sin^2 23^\circ & \sin^2 67^\circ & \cos 180^\circ \\ -\sin^2 67^\circ & -\sin^2 23^\circ & \cos^2 180^\circ \\ \cos 180^\circ & \sin^2 23^\circ & \sin^2 67^\circ \end{vmatrix} = 0.$$

State whether the statements in the Examples 15 to 18 is True or False.

Example 15: The determinant $\Delta = \begin{vmatrix} \cos(x+y) & -\sin(x+y) & \cos 2y \\ \sin x & \cos x & \sin y \\ -\cos x & \sin x & \cos y \end{vmatrix}$ is independent of x only.

$$\text{Ans: here, we have } \Delta = \begin{vmatrix} \cos(x+y) & -\sin(x+y) & \cos 2y \\ \sin x & \cos x & \sin y \\ -\cos x & \sin x & \cos y \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2 \sin y + R_3 \cos y$, we get

$$\Rightarrow \Delta = \begin{vmatrix} 0 & 0 & 1+\cos 2y \\ \sin x & \cos x & \sin y \\ -\cos x & \sin x & \cos y \end{vmatrix}$$

Now, expanding along R_1 , we get

$$\Rightarrow \Delta = (1+\cos 2y)(\sin^2 x + \cos^2 x)$$

$$\Rightarrow \Delta = (1 + \cos 2y)$$

Thus, value of given determinant is independent of x only.

Hence, given statement is True.

Example 16: The value of $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ {}^nC_1 & {}^{n+2}C_1 & {}^{n+4}C_1 \\ {}^nC_2 & {}^{n+2}C_2 & {}^{n+4}C_2 \end{vmatrix}$ is 8.

Ans: here, we have $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ {}^nC_1 & {}^{n+2}C_1 & {}^{n+4}C_1 \\ {}^nC_2 & {}^{n+2}C_2 & {}^{n+4}C_2 \end{vmatrix}$

$$\Rightarrow \Delta = \begin{vmatrix} 1 & 1 & 1 \\ \frac{n!}{(n-1)!} & \frac{(n+2)!}{(n+1)!} & \frac{(n+4)!}{(n+3)!} \\ \frac{n!}{(n-2)! 2!} & \frac{(n+2)!}{n! 2!} & \frac{(n+4)!}{(n+2)! 2!} \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} 1 & 1 & 1 \\ \frac{n(n-1)!}{(n-1)!} & \frac{(n+2)(n+1)!}{(n+1)!} & \frac{(n+4)(n+3)!}{(n+3)!} \\ \frac{n(n-1)(n-2)!}{(n-2)! 2!} & \frac{(n+2)(n+1)n!}{n! 2!} & \frac{(n+4)(n+3)(n+2)!}{(n+2)! 2!} \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} 1 & 1 & 1 \\ n & (n+2) & (n+4) \\ \frac{n(n-1)}{2} & \frac{(n+2)(n+1)}{2} & \frac{(n+4)(n+3)}{2} \end{vmatrix}$$

Taking $\frac{1}{2}$ as common from R_3 ,

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} 1 & 1 & 1 \\ n & (n+2) & (n+4) \\ n(n-1) & (n+2)(n+1) & (n+4)(n+3) \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 - C_3$ and $C_2 \rightarrow C_2 - C_3$, we get

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ -4 & -2 & (n+4) \\ -8n-12 & -4n-10 & (n+4)(n+3) \end{vmatrix}$$

Now, expanding along R_1 , we get

$$\Rightarrow \Delta = \frac{1}{2} [-4(-4n-10) + 2(-8n-12)]$$

$$\Rightarrow \Delta = \frac{1}{2} [16n+40-16n-24]$$

$$\Rightarrow \Delta = \frac{1}{2} \times 16$$

$$\Rightarrow \Delta = 8$$

Hence, the given statement is True.

Example 17: If $A = \begin{bmatrix} x & 5 & 2 \\ 2 & y & 3 \\ 1 & 1 & z \end{bmatrix}$, $xyz = 80$, $3x + 2y + 10z = 20$, then

$$A \text{ adj. } A = \begin{bmatrix} 81 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{bmatrix}.$$

Ans: here we have $A = \begin{bmatrix} x & 5 & 2 \\ 2 & y & 3 \\ 1 & 1 & z \end{bmatrix}$, $xyz=80$, $3x+2y+10z=20$

As we know that, $A^{-1} = \frac{1}{|A|} \text{adj}(A)$

$\Rightarrow AA^{-1} = \frac{1}{|A|} A \text{adj}(A)$

$\Rightarrow I = \frac{1}{|A|} A \text{adj}(A)$

$\Rightarrow A \text{adj}(A) = |A| \dots\dots\dots (i)$

Here, $|A| = x(yz-3) - 5(2z-3) + 2(2-y)$

$\Rightarrow |A| = xyz - 3x - 10z + 15 + 4 - 2y$

$\Rightarrow |A| = xyz + 19 - (3x + 2y + 10z)$

$\Rightarrow |A| = 80 + 19 - 20$

$\Rightarrow |A| = 79$

Therefore, $A \text{adj}(A) = 79$, and it is constant value.

Hence, the given statement is False.

Example 18: If $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} \frac{1}{2} & -4 & \frac{5}{2} \\ -\frac{1}{2} & 3 & -\frac{3}{2} \\ \frac{1}{2} & y & \frac{1}{2} \end{bmatrix}$, then $x = 1, y = -1$.

Ans: here, we have $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} \frac{1}{2} & -4 & \frac{5}{2} \\ -\frac{1}{2} & 3 & -\frac{3}{2} \\ \frac{1}{2} & y & \frac{1}{2} \end{bmatrix}$

As we know that, $AA^{-1} = I$

$$\Rightarrow \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -4 & \frac{5}{2} \\ -\frac{1}{2} & 3 & -\frac{3}{2} \\ \frac{1}{2} & y & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 - \frac{1}{2} + \frac{3}{2} & 0 + 3 + 3y & 0 - \frac{3}{2} + \frac{3}{2} \\ \frac{1}{2} - 1 + \frac{x}{2} & -4 + 6 + xy & \frac{5}{2} - 3 + \frac{x}{2} \\ 1 - 1 + 1 & -8 + 9 + y & 5 - \frac{9}{2} + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 3(1+y) & 0 \\ -\frac{1}{2} + \frac{x}{2} & 2 + xy & -\frac{1}{2} + \frac{x}{2} \\ 1 & 1 + y & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, comparing both sides, we get

$$\Rightarrow 3(1+y) = 0 \text{ and } -\frac{1}{2} + \frac{x}{2} = 0$$

$$\Rightarrow 1+y = 0 \text{ and } -1+x = 0$$

$$\Rightarrow y = -1 \text{ and } x = 1$$

Hence, the given statement is True.

Exercise 4.3

Using the properties of determinants in Exercise 1 to 6, evaluate

1. $\begin{vmatrix} x^2-x+1 & x-1 \\ x+1 & x+1 \end{vmatrix}$

Ans: here, we have $\begin{vmatrix} x^2-x+1 & x-1 \\ x+1 & x+1 \end{vmatrix}$

Applying $[C_1 \rightarrow C_1 - C_2]$, we get

$$\begin{aligned} &= \begin{vmatrix} x^2-2x+2 & x-1 \\ 0 & x+1 \end{vmatrix} \\ &= (x^2-2x+2)(x+1) - 0(x-1) \\ &= x^3 - 2x^2 + 2x + x^2 - 2x + 2 \\ &= x^3 - x^2 + 2 \end{aligned}$$

Hence, $\begin{vmatrix} x^2-x+1 & x-1 \\ x+1 & x+1 \end{vmatrix} = x^3 - x^2 + 2$.

2. $\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix}$

Ans: here, we have $\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix}$

Applying $[C_1 \rightarrow C_1 + C_2 + C_3]$, we get

$$= \begin{vmatrix} a+x+y+z & y & z \\ a+x+y+z & a+y & z \\ a+x+y+z & y & a+z \end{vmatrix}$$

Now, take common $(a + x + y + z)$ from C_1

$$= (a + x + y + z) \begin{vmatrix} 1 & y & z \\ 1 & a + y & z \\ 1 & y & a + z \end{vmatrix}$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$= (a + x + y + z) \begin{vmatrix} 1 & y & z \\ 0 & a & 0 \\ 0 & 0 & a \end{vmatrix}$$

Now, expanding along R_1

$$= (a + x + y + z) [(a \cdot a - 0) - y(0 - 0) + z(0 - 0)]$$

$$= (a + x + y + z) [a^2 - 0 - 0]$$

$$= a^2 (a + x + y + z)$$

Hence, $\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2 (a + x + y + z).$

$$3. \begin{vmatrix} 0 & xy^2 & xz^2 \\ x^2y & 0 & yz^2 \\ x^2z & zy^2 & 0 \end{vmatrix}$$

Ans: here, we have $\begin{vmatrix} 0 & xy^2 & xz^2 \\ x^2y & 0 & yz^2 \\ x^2z & zy^2 & 0 \end{vmatrix}$

Now, taking common x^2 , y^2 and z^2 from C_1 , C_2 and C_3 respectively.

$$= x^2 y^2 z^2 \begin{vmatrix} 0 & x & x \\ y & 0 & y \\ z & z & 0 \end{vmatrix}$$

Applying $[C_2 \rightarrow C_2 - C_3]$, we get

$$= x^2 y^2 z^2 \begin{vmatrix} 0 & 0 & x \\ y & -y & y \\ z & z & 0 \end{vmatrix}$$

Now, expanding along R_1

$$= x^2 y^2 z^2 [0(0 - yz) - 0(0 - zy) + x(yz + yz)]$$

$$= x^2 y^2 z^2 [x(yz + yz)]$$

$$= x^2 y^2 z^2 \times 2xyz$$

$$= 2x^3 y^3 z^3$$

Hence, $\begin{vmatrix} 0 & xy^2 & xz^2 \\ x^2 y & 0 & yz^2 \\ x^2 z & zy^2 & 0 \end{vmatrix} = 2x^3 y^3 z^3.$

4. $\begin{vmatrix} 3x & -x + y & -x + z \\ x - y & 3y & z - y \\ x - z & y - z & 3z \end{vmatrix}$

Ans: here, we have $\begin{vmatrix} 3x & -x + y & -x + z \\ x - y & 3y & z - y \\ x - z & y - z & 3z \end{vmatrix}$

Applying $[C_1 \rightarrow C_1 + C_2 + C_3]$, we get

$$= \begin{vmatrix} x + y + z & -x + y & -x + z \\ x + y + z & 3y & z - y \\ x + y + z & y - z & 3z \end{vmatrix}$$

Now, take common $(x + y + z)$ from C_1

$$= (x + y + z) \begin{vmatrix} 1 & -x + y & -x + z \\ 1 & 3y & z - y \\ 1 & y - z & 3z \end{vmatrix}$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$= (x + y + z) \begin{vmatrix} 1 & -x + y & -x + z \\ 0 & 2y + x & x - y \\ 0 & x - z & 2z + x \end{vmatrix}$$

Now, expanding along R_1

$$\begin{aligned} &= (x + y + z) [(2y + x)(2z + x) - (x - y)(x - z) - 0 + 0] \\ &= (x + y + z) [4yz + 2xy + 2xz + x^2 - (x^2 - xz - xy + yz)] \\ &= (x + y + z) [4yz + 2xy + 2xz + x^2 - x^2 + xz + xy - yz] \\ &= (x + y + z) [3yz + 3xy + 3xz] \\ &= 3(x + y + z)(yz + xy + xz) \end{aligned}$$

$$\text{Hence, } \begin{vmatrix} 3x & -x + y & -x + z \\ x - y & 3y & z - y \\ x - z & y - z & 3z \end{vmatrix} = 3(x + y + z)(yz + xy + xz).$$

$$5. \begin{vmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix}$$

$$\text{Ans: here, we have } \begin{vmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix}$$

Applying $[C_1 \rightarrow C_1 + C_2 + C_3]$, we get

$$= \begin{vmatrix} 3x+4 & x & x \\ 3x+4 & x+4 & x \\ 3x+4 & x & x+4 \end{vmatrix}$$

Now, taking common $(3x+4)$ from C_1

$$= (3x+4) \begin{vmatrix} 1 & x & x \\ 1 & x+4 & x \\ 1 & x & x+4 \end{vmatrix}$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$= (3x+4) \begin{vmatrix} 1 & x & x \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{vmatrix}$$

Now, expanding along R_1

$$= (3x+4) [(16-0)-0+0]$$

$$= 16(3x+4)$$

Hence, $\begin{vmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix} = 16(3x+4).$

6. $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$

Ans: here, we have $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$

Applying $[R_1 \rightarrow R_1 + R_2 + R_3]$, we get

$$= \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

Now, taking common $(a+b+c)$ from R_1

$$= (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

Applying $[C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1]$, we get

$$= (a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ 2b & -(a+b+c) & 0 \\ 2c & 0 & -(a+b+c) \end{vmatrix}$$

Now, expanding along R_1

$$= (a+b+c) [(a+b+c)^2 - 0]$$

$$= (a+b+c)^3$$

$$\text{Hence, } \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3.$$

Using the properties of determinants in Exercises 7 to 9.

$$7. \begin{vmatrix} y^2z^2 & yz & y+z \\ z^2x^2 & zx & z+x \\ x^2y^2 & xy & x+y \end{vmatrix} = 0$$

Ans: here, we have L.H.S =
$$\begin{vmatrix} y^2z^2 & yz & y+z \\ z^2x^2 & zx & z+x \\ x^2y^2 & xy & x+y \end{vmatrix}$$

Applying $[R_1 \rightarrow xR_1, R_2 \rightarrow yR_2 \text{ and } R_3 \rightarrow zR_3]$,

And it can be written as,

$$\Rightarrow \text{L.H.S} = \frac{1}{xyz} \begin{vmatrix} xy^2z^2 & xyz & xy+xz \\ yz^2x^2 & yzx & yz+xy \\ zx^2y^2 & zxy & zx+zy \end{vmatrix}$$

Taking common (xyz) from C_1 and C_2 , we get

$$\Rightarrow \text{L.H.S} = \frac{1}{xyz} \times (xyz)^2 \begin{vmatrix} yz & 1 & xy+xz \\ zx & 1 & yz+xy \\ xy & 1 & zx+zy \end{vmatrix}$$

$$\Rightarrow \text{L.H.S} = xyz \begin{vmatrix} yz & 1 & xy+xz \\ zx & 1 & yz+xy \\ xy & 1 & zx+zy \end{vmatrix}$$

Now, applying $[C_1 \rightarrow C_1 + C_3]$, we get

$$\Rightarrow \text{L.H.S} = xyz \begin{vmatrix} xy+yz+xz & 1 & xy+xz \\ xy+yz+xz & 1 & yz+xy \\ xy+yz+xz & 1 & zx+zy \end{vmatrix}$$

Taking common $(xy+yz+xz)$ from C_1

$$\Rightarrow \text{L.H.S} = xyz(xy+yz+xz) \begin{vmatrix} 1 & 1 & xy+xz \\ 1 & 1 & yz+xy \\ 1 & 1 & zx+zy \end{vmatrix}$$

$$\Rightarrow \text{L.H.S} = xyz(xy+yz+xz) \times 0$$

[when two columns are same, then value of determinants is 0]

$$\Rightarrow \text{L.H.S} = 0$$

$$\Rightarrow \text{L.H.S} = \text{R.H.S}$$

Hence Proved.

$$8. \begin{vmatrix} y+z & z & y \\ z & z+x & x \\ y & x & x+y \end{vmatrix} = 4xyz$$

Ans: here, we have L.H.S = $\begin{vmatrix} y+z & z & y \\ z & z+x & x \\ y & x & x+y \end{vmatrix}$

Applying $[C_1 \rightarrow C_1 + C_2 + C_3]$, we get

$$= \begin{vmatrix} 2(y+z) & z & y \\ 2(x+z) & z+x & x \\ 2(x+y) & x & x+y \end{vmatrix}$$

Now, taking common (2) from C_1

$$= 2 \begin{vmatrix} (y+z) & z & y \\ (x+z) & z+x & x \\ (x+y) & x & x+y \end{vmatrix}$$

Applying $[C_1 \rightarrow C_1 - C_2]$, we get

$$= 2 \begin{vmatrix} y & z & y \\ 0 & z+x & x \\ y & x & x+y \end{vmatrix}$$

Applying $[R_3 \rightarrow R_3 - R_1]$, we get

$$= 2 \begin{vmatrix} y & z & y \\ 0 & z+x & x \\ 0 & x-z & x \end{vmatrix}$$

Now, expanding along R_1

$$= 2 \left[y \{ xz + x^2 - (x^2 - xz) \} - 0 - 0 \right]$$

$$= 2 \left[y \{ xz + x^2 - x^2 + xz \} \right]$$

$$= 2 \times 2xyz$$

$$= 4xyz$$

$$= \text{R.H.S}$$

Thus, L.H.S = R.H.S

Hence proved.

$$9. \begin{vmatrix} a^2 + 2a & 2a + 1 & 1 \\ 2a + 1 & a + 2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a - 1)^3$$

$$\text{Ans: here, we have L.H.S} = \begin{vmatrix} a^2 + 2a & 2a + 1 & 1 \\ 2a + 1 & a + 2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$= \begin{vmatrix} a^2 + 2a & 2a + 1 & 1 \\ -(a^2 - 1) & -(a + 1) & 0 \\ -(a^2 + 2a - 3) & -(2a - 2) & 0 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 + 2a & 2a + 1 & 1 \\ -(a - 1)(a + 1) & -(a - 1) & 0 \\ -(a + 3)(a - 1) & -2(a - 1) & 0 \end{vmatrix}$$

Now, taking common $(a - 1)$ from R_2 and R_3

$$= (a - 1)^2 \begin{vmatrix} a^2 + 2a & 2a + 1 & 1 \\ -(a + 1) & -1 & 0 \\ -(a + 3) & -2 & 0 \end{vmatrix}$$

Now, expanding along R_1

$$= (a-1)^2 [0-0+1\{2a+2-(a+3)\}]$$

$$= (a-1)^2 (2a+2-a-3)$$

$$= (a-1)^2 (a-1)$$

$$= (a-1)^3$$

$$= \text{R.H.S}$$

Thus, L.H.S = R.H.S

Hence proved.

10. If $A+B+C=0$, then prove that
$$\begin{vmatrix} 1 & \cos C & \cos B \\ \cos C & 1 & \cos A \\ \cos B & \cos A & 1 \end{vmatrix} = 0.$$

Ans: here, we have $A+B+C=0$

$$\text{Now, L.H.S} = \begin{vmatrix} 1 & \cos C & \cos B \\ \cos C & 1 & \cos A \\ \cos B & \cos A & 1 \end{vmatrix}$$

Expanding along R_1 , we get

$$= 1(1-\cos^2 A) - \cos C(\cos C - \cos A \cdot \cos B) + \cos B(\cos A \cdot \cos C - \cos B)$$

$$= \sin^2 A - \cos^2 C + \cos A \cdot \cos B \cdot \cos C + \cos A \cdot \cos B \cdot \cos C - \cos^2 B$$

$$= \sin^2 A - \cos^2 B + 2\cos A \cdot \cos B \cdot \cos C - \cos^2 C$$

$$= -\cos(A+B) \cdot \cos(A-B) + 2\cos A \cdot \cos B \cdot \cos C - \cos^2 C$$

$$[\cos^2 B - \sin^2 A = \cos(A+B) \cdot \cos(A-B)]$$

$$= -\cos(-C) \cdot \cos(A-B) + \cos C(2\cos A \cdot \cos B - \cos C)$$

$$\begin{aligned}
 &= -\cos C(\cos A \cdot \cos B + \sin A \cdot \sin B) + \cos C(2 \cos A \cdot \cos B - \cos C) \\
 &= -\cos C(\cos A \cdot \cos B + \sin A \cdot \sin B - 2 \cos A \cdot \cos B + \cos C) \\
 &= -\cos C(\sin A \cdot \sin B - \cos A \cdot \cos B + \cos C) \\
 &= \cos C(\cos A \cdot \cos B - \sin A \cdot \sin B - \cos C) \\
 &= \cos C[\cos(A + B) - \cos C] \\
 &= \cos C[\cos(-C) - \cos C] \\
 &= \cos C[\cos C - \cos C] \\
 &= 0 \\
 &= \text{R.H.S}
 \end{aligned}$$

Thus, L.H.S = R.H.S

Hence proved.

11.If the co-ordinates of the vertices of an equilateral triangle with sides of length

'a' are $(x_1, y_1), (x_2, y_2), (x_3, y_3)$, then $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = \frac{3a^4}{4}$.

Ans: Since, we know that area of a triangle with vertices

$(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) is given by $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

$$\Rightarrow \Delta^2 = \frac{1}{4} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 \dots\dots\dots \text{eq (i)}$$

And we know that area of an equilateral triangle with side a,

$$\Rightarrow \Delta = \frac{\sqrt{3}}{4} \times a^2$$

$$\Rightarrow \Delta^2 = \left(\frac{\sqrt{3}}{4} \times a^2 \right)^2$$

$$\Rightarrow \Delta^2 = \frac{3}{16} \times a^4 \dots\dots\dots \text{Eq (ii)}$$

From eq (i) and (ii), we get

$$\Rightarrow \frac{1}{4} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = \frac{3}{16} \times a^4$$

$$\Rightarrow \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = \frac{3a^4}{4}$$

Hence proved.

12. Find the value of θ satisfying
$$\begin{vmatrix} 1 & 1 & \sin 3\theta \\ -4 & 3 & \cos 2\theta \\ 7 & -7 & -2 \end{vmatrix} = 0$$

Ans: here, we have
$$\begin{vmatrix} 1 & 1 & \sin 3\theta \\ -4 & 3 & \cos 2\theta \\ 7 & -7 & -2 \end{vmatrix} = 0$$

Applying $[C_1 \rightarrow C_1 - C_2]$, we get

$$\Rightarrow \begin{vmatrix} 0 & 1 & \sin 3\theta \\ -7 & 3 & \cos 2\theta \\ 14 & -7 & -2 \end{vmatrix} = 0$$

Taking common 7 from C_1

$$\Rightarrow 7 \begin{vmatrix} 0 & 1 & \sin 3\theta \\ -1 & 3 & \cos 2\theta \\ 2 & -7 & -2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 0 & 1 & \sin 3\theta \\ -1 & 3 & \cos 2\theta \\ 2 & -7 & -2 \end{vmatrix} = 0$$

Now, expanding along R_1

$$\Rightarrow 0(-6 + 7 \cos 2\theta) - 1(2 - 2 \cos 2\theta) + \sin 3\theta(7 - 6) = 0$$

$$\Rightarrow -2 + 2 \cos 2\theta + \sin 3\theta = 0$$

$$\Rightarrow -2 + 2(1 - 2 \sin^2 \theta) + 3 \sin \theta - 4 \sin^3 \theta = 0$$

$$\Rightarrow -2 + 2 - 4 \sin^2 \theta + 3 \sin \theta - 4 \sin^3 \theta = 0$$

$$\Rightarrow 4 \sin^3 \theta + 4 \sin^2 \theta - 3 \sin \theta = 0$$

$$\Rightarrow \sin \theta (4 \sin^2 \theta + 4 \sin \theta - 3) = 0$$

$$\Rightarrow \sin \theta (4 \sin^2 \theta + 6 \sin \theta - 2 \sin \theta - 3) = 0$$

$$\Rightarrow \sin \theta \{ 2 \sin \theta (2 \sin \theta + 3) - 1(2 \sin \theta + 3) \} = 0$$

$$\Rightarrow \sin \theta (2 \sin \theta + 3)(2 \sin \theta - 1) = 0$$

$$\Rightarrow \sin \theta = 0, \sin \theta = -\frac{3}{2}, \sin \theta = \frac{1}{2}$$

Now, Case 1: When $\sin \theta = 0$

$$\Rightarrow \sin \theta = \sin 0^\circ$$

$$\Rightarrow \theta = n\pi, \text{ where } n \in \mathbb{Z}$$

Case 2: When $\sin \theta = -\frac{3}{2}$

Here, it is not possible because $\sin \theta \in [-1, 1]$

Case 3: when $\sin \theta = \frac{1}{2}$

$$\Rightarrow \sin \theta = \sin \frac{\pi}{6}$$

$$\Rightarrow \sin \theta = \sin \frac{\pi}{6}$$

$$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{6}, \text{ where } n \in \mathbb{Z}$$

Hence, values of θ are $n\pi$ or $n\pi + (-1)^n \frac{\pi}{6}$, where $n \in \mathbb{Z}$.

13. If $\begin{vmatrix} 4-x & 4+x & 4+x \\ 4+x & 4-x & 4+x \\ 4+x & 4+x & 4-x \end{vmatrix} = 0$, then find the value of x .

Ans: here, we have $\begin{vmatrix} 4-x & 4+x & 4+x \\ 4+x & 4-x & 4+x \\ 4+x & 4+x & 4-x \end{vmatrix} = 0$

Applying $[C_1 \rightarrow C_1 + C_2 + C_3]$, we get

$$\Rightarrow \begin{vmatrix} 12+x & 4+x & 4+x \\ 12+x & 4-x & 4+x \\ 12+x & 4+x & 4-x \end{vmatrix} = 0$$

Now, taking common $(12+x)$ from C_1

$$\Rightarrow (12+x) \begin{vmatrix} 1 & 4+x & 4+x \\ 1 & 4-x & 4+x \\ 1 & 4+x & 4-x \end{vmatrix} = 0$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$\Rightarrow (12+x) \begin{vmatrix} 1 & 4+x & 4+x \\ 0 & -2x & 0 \\ 0 & 0 & -2x \end{vmatrix} = 0$$

Now, expanding along C_1

$$\Rightarrow (12+x)(4x^2) = 0$$

$$\Rightarrow (12+x) = 0 \text{ or } (4x^2) = 0$$

$$\Rightarrow x = -12 \text{ or } x = 0$$

Hence, Value of x are -12 and 0 .

14.If $a_1, a_2, a_3, \dots, a_r$ are in G.P, then prove that the determinant

$$\begin{vmatrix} a_{r+1} & a_{r+5} & a_{r+9} \\ a_{r+7} & a_{r+11} & a_{r+15} \\ a_{r+11} & a_{r+17} & a_{r+21} \end{vmatrix} \text{ is independent of } r.$$

Ans: Let the first term of given G.P be A and common ratio be R .

Then, n th term, $a_n = AR^{n-1}$

Therefore,

$$\begin{aligned} &= \begin{vmatrix} a_{r+1} & a_{r+5} & a_{r+9} \\ a_{r+7} & a_{r+11} & a_{r+15} \\ a_{r+11} & a_{r+17} & a_{r+21} \end{vmatrix} \\ &= \begin{vmatrix} AR^r & AR^{r+4} & AR^{r+8} \\ AR^{r+6} & AR^{r+10} & AR^{r+14} \\ AR^{r+10} & AR^{r+16} & AR^{r+20} \end{vmatrix} \end{aligned}$$

Taking common AR^r , AR^{r+6} and AR^{r+10} from R_1 , R_2 and R_3 respectively.

$$= AR^r \cdot AR^{r+6} \cdot AR^{r+10} \begin{vmatrix} 1 & R^4 & R^8 \\ 1 & R^4 & R^8 \\ 1 & R^6 & R^{10} \end{vmatrix}$$

$= 0$ [two rows are identical, hence value of determinant is 0]

Hence, the determinant $\begin{vmatrix} a_{r+1} & a_{r+5} & a_{r+9} \\ a_{r+7} & a_{r+11} & a_{r+15} \\ a_{r+11} & a_{r+17} & a_{r+21} \end{vmatrix}$ is independent of r .

15. Show that the points $(a+5, a-4)$, $(a-2, a+3)$ and (a, a) do not lie on a straight line for any value of a .

Ans: here, given points are $(a+5, a-4)$, $(a-2, a+3)$ and (a, a) .

$$\text{Now, } \Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} a+5 & a-4 & 1 \\ a-2 & a+3 & 1 \\ a & a & 1 \end{vmatrix}$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} a+5 & a-4 & 1 \\ -7 & 7 & 0 \\ -5 & 4 & 0 \end{vmatrix}$$

Now, expanding along C_3

$$\Rightarrow \Delta = \frac{1}{2} [-28 + 35]$$

$$\Rightarrow \Delta = \frac{7}{2} \text{ sq. units}$$

$$\Rightarrow \Delta \neq 0$$

Here, area of this triangle is not equal to 0.

Hence, given points form a triangle i.e., points do not lie in a straight line.

16. Show that the ΔABC is an isosceles triangle if the determinant,

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos^2 A + \cos A & \cos^2 B + \cos B & \cos^2 C + \cos C \end{vmatrix} = 0$$

Ans: here, we have $\begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos^2 A + \cos A & \cos^2 B + \cos B & \cos^2 C + \cos C \end{vmatrix} = 0$

Now, applying $[C_2 \rightarrow C_2 - C_1 \text{ \& } C_3 \rightarrow C_3 - C_1]$, we get

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos^2 A + \cos A & \cos^2 B + \cos B & \cos^2 C + \cos C \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 1 + \cos A & \cos B - \cos A & \cos C - \cos A \\ \cos^2 A + \cos A & \cos^2 B - \cos^2 A + \cos B - \cos A & \cos^2 C - \cos^2 A + \cos C - \cos A \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 1 + \cos A & \cos B - \cos A & \cos C - \cos A \\ \cos^2 A + \cos A & (\cos B - \cos A)(\cos B + \cos A + 1) & (\cos C - \cos A)(\cos C + \cos A + 1) \end{vmatrix} = 0$$

Now, take common $(\cos B - \cos A)$ and $(\cos C - \cos A)$ from C_1 and C_2 respectively, we get

$$\Rightarrow (\cos B - \cos A)(\cos C - \cos A) \begin{vmatrix} 1 & 0 & 0 \\ 1 + \cos A & 1 & 1 \\ \cos^2 A + \cos A & (\cos B + \cos A + 1) & (\cos C + \cos A + 1) \end{vmatrix} = 0$$

Expanding along R_1 ,

$$\Rightarrow (\cos B - \cos A)(\cos C - \cos A)(\cos C + \cos A + 1 - \cos B - \cos A - 1) = 0$$

$$\Rightarrow (\cos B - \cos A)(\cos C - \cos A)(\cos C - \cos B) = 0$$

$$\Rightarrow (\cos B - \cos A) = 0 \text{ or } (\cos C - \cos A) = 0 \text{ or } (\cos C - \cos B) = 0$$

$$\Rightarrow \cos B = \cos A \text{ or } \cos C = \cos A \text{ or } \cos C = \cos B$$

$$\Rightarrow B = A \text{ or } C = A \text{ or } C = B$$

Hence, ΔABC is an isosceles triangle.

17. Find A^{-1} if $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and show that $A^{-1} = \frac{A^2 - 3I}{2}$.

Ans: here, we have $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

Here, minors of elements are

$$\Rightarrow M_{11} = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1, \quad M_{12} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1, \quad M_{13} = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1$$

$$\Rightarrow M_{21} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1, \quad M_{22} = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1, \quad M_{23} = \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1$$

$$\Rightarrow M_{31} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1, \quad M_{32} = \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1, \quad M_{33} = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$$

Now, cofactor of an element a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = -1, \quad A_{12} = 1, \quad A_{13} = 1$$

$$\Rightarrow A_{21} = 1, \quad A_{22} = -1, \quad A_{23} = 1$$

$$\Rightarrow A_{31} = 1, \quad A_{32} = 1, \quad A_{33} = -1$$

Thus, matrix formed by cofactors, say $P = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$

Now, Adjoint matrix, $\text{adj } A = \text{transpose of matrix } P$

$$\Rightarrow \text{adj } A = [P]^T$$

$$\Rightarrow \text{adj } A = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}^T$$

$$\Rightarrow \text{adj } A = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

$$\text{And } |A| = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$\Rightarrow |A| = 0(0-1) - 1(0-1) + 1(1-0)$$

$$\Rightarrow |A| = 1 + 1 = 2$$

$$\text{Now, L.H.S} = A^{-1}$$

$$= \frac{\text{adj } A}{|A|}$$

$$= \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

$$\text{And R.H.S} = \frac{A^2 - 3I}{2}$$

$$= \frac{1}{2} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

Thus, L.H.S = R.H.S

Hence proved.

Long Answer (L.A)

18. If $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, find A^{-1} . Using A^{-1} , solve the system of linear equations $x - 2y = 10$, $2x - y - z = 8$, $-2y + z = 7$.

Ans: here, we have $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$ Eq(i)

$$\text{Now, } |A| = \begin{vmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{vmatrix}$$

$$\Rightarrow |A| = 1(-1-2) - 2(-2-0) + 0(2-0)$$

$$\Rightarrow |A| = -3 + 4 = 1$$

Here, minors of elements are

$$\Rightarrow M_{11} = \begin{vmatrix} -1 & -2 \\ -1 & 1 \end{vmatrix} = -3, M_{12} = \begin{vmatrix} -2 & -2 \\ 0 & 1 \end{vmatrix} = -2, M_{13} = \begin{vmatrix} -2 & -1 \\ 0 & -1 \end{vmatrix} = 2$$

$$\Rightarrow M_{21} = \begin{vmatrix} 2 & 0 \\ -1 & 1 \end{vmatrix} = 2, M_{22} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1, M_{23} = \begin{vmatrix} 1 & 2 \\ 0 & -1 \end{vmatrix} = -1$$

$$\Rightarrow M_{31} = \begin{vmatrix} 2 & 0 \\ -1 & -2 \end{vmatrix} = -4, M_{32} = \begin{vmatrix} 1 & 0 \\ -2 & -2 \end{vmatrix} = -2, M_{33} = \begin{vmatrix} 1 & 2 \\ -2 & -1 \end{vmatrix} = 3$$

Now, cofactor of an element a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = -3, A_{12} = 2, A_{13} = 2$$

$$\Rightarrow A_{21} = -2, A_{22} = 1, A_{23} = 1$$

$$\Rightarrow A_{31} = -4, A_{32} = 2, A_{33} = 3$$

Thus, matrix formed by cofactors, say $P = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix}$

Now, Adjoint matrix, $\text{adj } A = \text{transpose of matrix } P$

$$\Rightarrow \text{adj } A = [P]^T$$

$$\Rightarrow \text{adj } A = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix}^T$$

$$\Rightarrow \text{adj } A = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

Now, $A^{-1} = \frac{\text{adj } A}{|A|}$

$$\Rightarrow A^{-1} = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \dots\dots\dots \text{eq(ii)}$$

Hence, the required A^{-1} is $\begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$.

Also, we have the system of linear equations as

$$x - 2y = 10, \quad 2x - y - z = 8, \quad -2y + z = 7$$

These equations can be written as,

$$\begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$$

Let $C = \begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ & $D = \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$

$$\Rightarrow CX = D$$

We know that, $(A^T)^{-1} = (A^{-1})^T$

$$\Rightarrow C^T = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix} = A \quad [\text{from eq (i)}]$$

$$\Rightarrow X = C^{-1}D = (A^{-1})^T D$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -30 + 16 + 14 \\ -20 + 8 + 7 \\ -40 + 16 + 21 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ -3 \end{bmatrix}$$

On comparing, we get

$$\Rightarrow x = 0, y = -5, z = -3$$

Hence, values of x , y and z are 0, -5 and -3 respectively.

19. Using matrix method, solve the system of equations $3x+2y-2z=3$, $x+2y+3z=6$, $2x-y+z=2$.

Ans: here, we have $3x + 2y - 2z = 3$, $x + 2y + 3z = 6$, $2x - y + z = 2$

These, equations can be written as,

$$\Rightarrow \begin{bmatrix} 3 & 2 & -2 \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$$

$$\text{Let } A = \begin{bmatrix} 3 & 2 & -2 \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ \& } B = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$$

$$\Rightarrow AX = B$$

$$\Rightarrow X = A^{-1}B \dots\dots\dots \text{eq (i),}$$

$$\text{Now, } |A| = \begin{vmatrix} 3 & 2 & -2 \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{vmatrix}$$

$$\Rightarrow |A| = 3(2+3) - 2(1-6) - 2(-1-4)$$

$$\Rightarrow |A| = 15 + 10 + 10$$

$$\Rightarrow |A| = 35$$

Here, minors of elements are

$$\Rightarrow M_{11} = \begin{vmatrix} 2 & 3 \\ -1 & 1 \end{vmatrix} = 5, M_{12} = \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} = -5, M_{13} = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = -5$$

$$\Rightarrow M_{21} = \begin{vmatrix} 2 & -2 \\ -1 & 1 \end{vmatrix} = 0, M_{22} = \begin{vmatrix} 3 & -2 \\ 2 & 1 \end{vmatrix} = 7, M_{23} = \begin{vmatrix} 3 & 2 \\ 2 & -1 \end{vmatrix} = -7$$

$$\Rightarrow M_{31} = \begin{vmatrix} 2 & -2 \\ 2 & 3 \end{vmatrix} = 10, M_{32} = \begin{vmatrix} 3 & -2 \\ 1 & 3 \end{vmatrix} = 11, M_{33} = \begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} = 4$$

Now, cofactor of an element a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.

$$\Rightarrow A_{11} = (-1)^{1+1} M_{11} = 5, A_{12} = 5, A_{13} = -5$$

$$\Rightarrow A_{21} = 0, A_{22} = 7, A_{23} = 7$$

$$\Rightarrow A_{31} = 10, A_{32} = -11, A_{33} = 4$$

Thus, matrix formed by cofactors, say $P = \begin{bmatrix} 5 & 5 & -5 \\ 0 & 7 & 7 \\ 10 & -11 & 4 \end{bmatrix}$

Now, Adjoint matrix, $\text{adj } A = \text{transpose of matrix } P$

$$\Rightarrow \text{adj } A = [P]^T$$

$$\Rightarrow \text{adj } A = \begin{bmatrix} 5 & 5 & -5 \\ 0 & 7 & 7 \\ 10 & -11 & 4 \end{bmatrix}^T$$

$$\Rightarrow \text{adj } A = \begin{bmatrix} 5 & 0 & 10 \\ 5 & 7 & -11 \\ -5 & 7 & 4 \end{bmatrix}$$

$$\text{Now, } A^{-1} = \frac{\text{adj } A}{|A|}$$

$$\Rightarrow A^{-1} = \frac{1}{35} \begin{bmatrix} 5 & 0 & 10 \\ 5 & 7 & -11 \\ -5 & 7 & 4 \end{bmatrix}$$

$$\text{Now, } X = A^{-1}B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{35} \begin{bmatrix} 5 & 0 & 10 \\ 5 & 7 & -11 \\ -5 & 7 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{35} \begin{bmatrix} 15 + 0 + 20 \\ 15 + 42 - 22 \\ -15 + 42 + 8 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{35} \begin{bmatrix} 35 \\ 35 \\ 35 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{35}{35} \\ \frac{35}{35} \\ \frac{35}{35} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

On comparing, we get

$$\Rightarrow x = 1, y = 1, z = 1$$

Hence, values of x, y and z are 1, 1 and 1 respectively.

20. Given $A = \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$, find BA and use this to solve the system of equations $y+2z=7$, $x-y=3$, $2x+3y+4z=17$.

Ans: here, we have $A = \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$

$$\Rightarrow BA = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$$

$$\Rightarrow BA = \begin{bmatrix} 2+4+0 & 2-2+0 & -4+4+0 \\ 4-12+8 & 4+6-4 & -8-12+20 \\ 0-4+4 & 0+2-2 & 0-4+10 \end{bmatrix}$$

$$\Rightarrow BA = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$\Rightarrow BA = 6 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow BA = 6I$$

$$\Rightarrow B^{-1} = \frac{A}{6}$$

$$\Rightarrow B^{-1} = \frac{1}{6} \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix} \dots\dots\dots \text{Eq (i)}$$

Also, we have $x - y = 3$, $2x + 3y + 4z = 17$, $y + 2z = 7$

These, equations can be written as,

$$\Rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix} \quad [\text{from eq (i)}]$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 6 + 34 - 28 \\ -12 + 34 - 28 \\ 6 - 17 + 35 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 12 \\ -6 \\ 24 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{12}{6} \\ -\frac{6}{6} \\ \frac{24}{6} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

On comparing, we get

$$\Rightarrow x = 2, y = -1, z = 4$$

Hence, values of x, y and z are 2, -1 and 4 respectively.

21. If $a + b + c \neq 0$ and $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$, then prove that $a = b = c$.

Ans: here, we have $a + b + c \neq 0$ and $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$$\Rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

Applying $[C_1 \rightarrow C_1 + C_2 + C_3]$, we get

$$\Rightarrow \begin{vmatrix} a+b+c & b & c \\ a+b+c & c & a \\ a+b+c & a & b \end{vmatrix} = 0$$

Now, taking common $(a + b + c)$ from C_1

$$\Rightarrow (a + b + c) \begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix} = 0$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$\Rightarrow (a + b + c) \begin{vmatrix} 1 & b & c \\ 0 & c-b & a-c \\ 0 & a-b & b-c \end{vmatrix} = 0$$

Now, expanding along C_1

$$\Rightarrow (a + b + c) [(c-b)(b-c) - (a-b)(a-c)] = 0$$

$$\Rightarrow (a + b + c) [bc - c^2 - b^2 + bc - (a^2 - ac - ab + bc)] = 0$$

$$\Rightarrow (a + b + c) (2bc - c^2 - b^2 - a^2 + ac + ab - bc) = 0$$

$$\Rightarrow (a+b+c)(bc-c^2-b^2-a^2+ac+ab)=0$$

$$\Rightarrow -(a+b+c)(a^2+b^2+c^2-ab-bc-ac)=0$$

$$\Rightarrow -\frac{1}{2}(a+b+c)(2a^2+2b^2+2c^2-2ab-2bc-2ac)=0$$

$$\Rightarrow (a+b+c)[a^2+a^2+b^2+b^2+c^2+c^2-2ab-2bc-2ac]=0$$

$$\Rightarrow (a+b+c)[(a^2+b^2-2ab)+(b^2+c^2-2bc)+c^2+a^2-2ac]=0$$

$$\Rightarrow (a+b+c)[(a-b)^2+(b-c)^2+(c-a)^2]=0$$

Here, $(a+b+c) \neq 0$

Therefore,

$$\Rightarrow (a-b)^2+(b-c)^2+(c-a)^2=0$$

It is possible when $(a-b)^2=0$, $(b-c)^2=0$, $(c-a)^2=0$

$$\Rightarrow a-b=0, b-c=0, c-a=0$$

$$\Rightarrow a=b, b=c, c=a$$

Thus, $a=b=c$.

Hence proved.

22. Prove that $\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$ is divisible by $(a+b+c)$ and find the quotient.

Ans: here, we have $\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$

Applying $[C_1 \rightarrow C_1 - C_2]$ & $[C_2 \rightarrow C_2 - C_3]$, we get

$$= \begin{vmatrix} bc - a^2 - ca + b^2 & ca - b^2 - ab + c^2 & ab - c^2 \\ ca - b^2 - ab + c^2 & ab - c^2 - bc + a^2 & bc - a^2 \\ ab - c^2 - bc + a^2 & bc - a^2 - ca + b^2 & ca - b^2 \end{vmatrix}$$

$$= \begin{vmatrix} (b-a)(a+b+c) & (c-b)(a+b+c) & ab - c^2 \\ (c-b)(a+b+c) & (a-c)(a+b+c) & bc - a^2 \\ (a-c)(a+b+c) & (b-a)(a+b+c) & ca - b^2 \end{vmatrix}$$

Taking $(a+b+c)$ common from C_1 and C_2

$$= (a+b+c)^2 \begin{vmatrix} (b-a) & (c-b) & ab - c^2 \\ (c-b) & (a-c) & bc - a^2 \\ (a-c) & (b-a) & ca - b^2 \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$

$$= (a+b+c)^2 \begin{vmatrix} 0 & 0 & ab + bc + ca - a^2 - b^2 - c^2 \\ (c-b) & (a-c) & bc - a^2 \\ (a-c) & (b-a) & ca - b^2 \end{vmatrix}$$

Taking common $(ab + bc + ca - a^2 - b^2 - c^2)$ from R_1

$$= (a+b+c)^2 (ab + bc + ca - a^2 - b^2 - c^2) \begin{vmatrix} 0 & 0 & 1 \\ (c-b) & (a-c) & bc - a^2 \\ (a-c) & (b-a) & ca - b^2 \end{vmatrix}$$

$$= -(a+b+c)^2 (a^2 + b^2 + c^2 - ab - bc - ac) \begin{vmatrix} 0 & 0 & 1 \\ (c-b) & (a-c) & bc - a^2 \\ (a-c) & (b-a) & ca - b^2 \end{vmatrix}$$

$$= -(a+b+c)(a^3 + b^3 + c^3 - 3abc) \begin{vmatrix} 0 & 0 & 1 \\ (c-b) & (a-c) & bc - a^2 \\ (a-c) & (b-a) & ca - b^2 \end{vmatrix}$$

Expanding along R_1

$$= -(a+b+c)(a^3 + b^3 + c^3 - 3abc)[(c-b)(b-a) - (a-c)(a-c)]$$

$$\begin{aligned}
 &= -(a+b+c)(a^3+b^3+c^3-3abc)[bc-ac-b^2+ab-(a^2+c^2-2ac)] \\
 &= -(a+b+c)(a^3+b^3+c^3-3abc)(bc-ac-b^2+ab-a^2-c^2+2ac) \\
 &= -(a+b+c)(a^3+b^3+c^3-3abc)(-a^2-b^2-c^2+ab+bc+ac) \\
 &= \frac{1}{2}(a+b+c)(a^3+b^3+c^3-3abc)[(a-b)^2+(b-c)^2+(c-a)^2]
 \end{aligned}$$

Hence, given determinant is divisible by $(a+b+c)$.

And quotient is $\frac{1}{2}(a^3+b^3+c^3-3abc)[(a-b)^2+(b-c)^2+(c-a)^2]$.

23. If $x+y+z=0$, prove that
$$\begin{vmatrix} xa & yb & zc \\ yc & za & xb \\ zb & xc & ya \end{vmatrix} = xyz \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}.$$

Ans: here, we have $x+y+z=0$

Now, L.H.S =
$$\begin{vmatrix} xa & yb & zc \\ yc & za & xb \\ zb & xc & ya \end{vmatrix}$$

Expanding along R_1

$$= xa(a^2yz - x^2bc) - yb(y^2ac - b^2xz) + zc(c^2xy - z^2ab)$$

$$= a^3xyz - x^3abc - y^3abc + b^3xyz + c^3xyz - z^3abc$$

$$= xyz(a^3 + b^3 + c^3) - abc(x^3 + y^3 + z^3)$$

$$= xyz(a^3 + b^3 + c^3) - abc \times 3xyz$$

$$[\text{if } x+y+z=0, x^3+y^3+z^3=3xyz]$$

$$= xyz(a^3 + b^3 + c^3 - 3abc)$$

$$\text{Now, R.H.S} = xyz \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$$

Applying $[C_1 \rightarrow C_1 + C_2 + C_3]$, we get

$$= xyz \begin{vmatrix} a+b+c & b & c \\ a+b+c & a & b \\ a+b+c & c & a \end{vmatrix}$$

Taking common $(a+b+c)$ from C_1

$$= xyz(a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & a & b \\ 1 & c & a \end{vmatrix}$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$= xyz(a+b+c) \begin{vmatrix} 1 & b & c \\ 0 & a-b & b-c \\ 0 & c-b & a-c \end{vmatrix}$$

Now, expanding along C_1 , we get

$$\begin{aligned} &= xyz(a+b+c) [(a-b)(a-c) - (c-b)(b-c)] \\ &= xyz(a+b+c) [a^2 - ac - ab + bc - (bc - c^2 - b^2 + bc)] \\ &= xyz(a+b+c) (a^2 - ac - ab + bc - bc + c^2 + b^2 - bc) \\ &= xyz(a+b+c) (a^2 + b^2 + c^2 - ab - bc - ac) \\ &= xyz(a^3 + b^3 + c^3 - 3abc) \end{aligned}$$

Thus, L.H.S = R.H.S

Hence proved.

Objective Type Questions (M.C.Q)

Choose the correct answer from given four options in each of the exercise from 24 to 37.

24. If $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$, then value of x is

(A) 3

(B) ± 3

(C) ± 6

(D) 6

Ans: here, we have $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$

$$\Rightarrow 2x^2 - 40 = 18 + 14$$

$$\Rightarrow 2x^2 - 40 = 32$$

$$\Rightarrow 2x^2 = 72$$

$$\Rightarrow x^2 = 36$$

$$\Rightarrow x = \sqrt{36}$$

$$\Rightarrow x = \pm 6$$

Hence, option (C) is correct answer.

25. The value of determinant $\begin{vmatrix} a-b & b+c & a \\ b-a & c+a & b \\ c-a & a+b & c \end{vmatrix}$

(A) $a^3 + b^3 + c^3$

(B) $3bc$

(C) $a^3 + b^3 + c^3 - 3abc$

(D) none of these

Ans: here, we have
$$\begin{vmatrix} a-b & b+c & a \\ b-a & c+a & b \\ c-a & a+b & c \end{vmatrix}$$

Applying $C_3 \rightarrow C_2 + C_3$

$$= \begin{vmatrix} a-b & b+c & a \\ b-a & c+a & b \\ c-a & a+b & c \end{vmatrix}$$

$$= \begin{vmatrix} a-b & b+c & a+b+c \\ b-a & c+a & a+b+c \\ c-a & a+b & a+b+c \end{vmatrix}$$

Taking common $(a+b+c)$ from C_3 , we get

$$= (a+b+c) \begin{vmatrix} a-b & b+c & 1 \\ b-a & c+a & 1 \\ c-a & a+b & 1 \end{vmatrix}$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$= (a+b+c) \begin{vmatrix} a-b & b+c & 1 \\ -2(a-b) & a-b & 0 \\ c-2a+b & a-c & 0 \end{vmatrix}$$

Taking common $(a-b)$ from R_2

$$= (a+b+c)(a-b) \begin{vmatrix} a-b & b+c & 1 \\ -2 & 1 & 0 \\ c-2a+b & a-c & 0 \end{vmatrix}$$

Expanding along C_3

$$= (a+b+c)(a-b)(-2a+2c-c+2a-b)$$

$$= (a + b + c)(a - b)(c - b)$$

Hence, option (D) is correct.

26. The area of a triangle with the vertices $(-3, 0)$, $(3, 0)$ and $(0, k)$ is 9 sq. units. The value of k will be

(A) 9

(B) 3

(C) -9

(D) 6

Ans: we know that, area of a triangle with vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is given by

$$\Rightarrow \Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$\Rightarrow 9 = \frac{1}{2} \begin{vmatrix} -3 & 0 & 1 \\ 3 & 0 & 1 \\ 0 & k & 1 \end{vmatrix}$$

$$\Rightarrow 18 = \begin{vmatrix} -3 & 0 & 1 \\ 3 & 0 & 1 \\ 0 & k & 1 \end{vmatrix}$$

Expanding along C_2

$$\Rightarrow 18 = -k(-3-3)$$

$$\Rightarrow 18 = 6k$$

$$\Rightarrow \frac{18}{6} = k$$

$$\Rightarrow 3 = k$$

Thus, value of k is 3.

Hence, option (B) is correct.

27. The determinant $\begin{vmatrix} b^2 - ab & b - c & bc - ac \\ ab - a^2 & a - b & b^2 - ab \\ bc - ac & c - a & ab - a^2 \end{vmatrix}$ equals

(A) $abc(b - c)(c - a)(a - b)$

(B) $(b - c)(c - a)(a - b)$

(C) $(a + b + c)(b - c)(c - a)(a - b)$

(D) None of these

Ans: here, we have $\begin{vmatrix} b^2 - ab & b - c & bc - ac \\ ab - a^2 & a - b & b^2 - ab \\ bc - ac & c - a & ab - a^2 \end{vmatrix}$

$$= \begin{vmatrix} b(b - a) & b - c & c(b - a) \\ a(b - a) & a - b & b(b - a) \\ c(b - a) & c - a & a(b - a) \end{vmatrix}$$

Taking common $(b - a)$ from C_1 and C_3

$$= (b - a)^2 \begin{vmatrix} b & b - c & c \\ a & a - b & b \\ c & c - a & a \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 + C_3$

$$= (b - a)^2 \begin{vmatrix} b & b & c \\ a & a & b \\ c & c & a \end{vmatrix} = 0$$

Here, two columns of determinant are identical. Thus, value of this determinant is 0.

Hence, option (D) is correct answer.

28. The number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval

$$-\frac{\pi}{4} \leq x \leq \frac{\pi}{4} \text{ is}$$

(A) 0

(B) 2

(C) 1

(D) 3

Ans: here, we have $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$

Applying $[C_1 \rightarrow C_1 + C_2 + C_3]$, we get

$$\Rightarrow \begin{vmatrix} 2\cos x + \sin x & \cos x & \cos x \\ 2\cos x + \sin x & \sin x & \cos x \\ 2\cos x + \sin x & \cos x & \sin x \end{vmatrix} = 0$$

Taking common $(2\cos x + \sin x)$ from C_1

$$\Rightarrow (2\cos x + \sin x) \begin{vmatrix} 1 & \cos x & \cos x \\ 1 & \sin x & \cos x \\ 1 & \cos x & \sin x \end{vmatrix} = 0$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$\Rightarrow (2\cos x + \sin x) \begin{vmatrix} 1 & \cos x & \cos x \\ 0 & \sin x - \cos x & 0 \\ 0 & 0 & \sin x - \cos x \end{vmatrix} = 0$$

Expanding along C_1

$$\Rightarrow (2\cos x + \sin x)(\sin x - \cos x)^2 = 0$$

$$\Rightarrow (2\cos x + \sin x) = 0 \text{ or } (\sin x - \cos x)^2 = 0$$

$$\Rightarrow \sin x = -2\cos x \text{ or } \sin x - \cos x = 0$$

$$\Rightarrow \tan x = -2 \text{ or } \sin x = \cos x$$

$$\Rightarrow \tan x = -2 \text{ or } \tan x = 1$$

$$\text{Also, we have } -\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$$

$$\Rightarrow -\tan \frac{\pi}{4} \leq \tan x \leq \tan \frac{\pi}{4}$$

$$\Rightarrow -1 \leq \tan x \leq 1$$

Thus, $\tan x = -2$ is not possible.

Therefore, $\tan x = 1$

$$\Rightarrow x = \frac{\pi}{4}$$

So, only one distinct real root exists.

Hence, option (C) is correct answer.

29. If A, B and C are angles of a triangle, then the determinant

$$\begin{vmatrix} -1 & \cos C & \cos B \\ \cos C & -1 & \cos A \\ \cos B & \cos A & -1 \end{vmatrix} \text{ is equal to}$$

(A) 0

(B) -1

(C) 1

(D) None of these

Ans: Here, we have
$$\begin{vmatrix} -1 & \cos C & \cos B \\ \cos C & -1 & \cos A \\ \cos B & \cos A & -1 \end{vmatrix}$$

Applying $[C_1 \rightarrow aC_1 + bC_2 + cC_3]$, we get

$$= \begin{vmatrix} -a + b.\cos C + c.\cos B & \cos C & \cos B \\ a \cos C - b + c.\cos A & -1 & \cos A \\ a.\cos B + b.\cos A - c & \cos A & -1 \end{vmatrix}$$

Also, by projection rule in a triangle, we know that

$$\Rightarrow a = b.\cos C + c.\cos B$$

$$\Rightarrow b = a \cos C + c.\cos A$$

$$\Rightarrow c = a.\cos B + b.\cos A$$

Using above equations in column first, we get

$$= \begin{vmatrix} -a + a & \cos C & \cos B \\ b - b & -1 & \cos A \\ c - c & \cos A & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & \cos C & \cos B \\ 0 & -1 & \cos A \\ 0 & \cos A & -1 \end{vmatrix} = 0$$

Since, determinant having all elements of any column or row gives value of determinant is zero.

Hence, option (A) is correct answer.

30. Let $f(t) = \begin{vmatrix} \cos t & t & 1 \\ 2 \sin t & t & 2t \\ \sin t & t & t \end{vmatrix}$, then $\lim_{t \rightarrow 0} \frac{f(t)}{t^2}$ is equal to

(A) 0

(B) -1

(C) 2

(D) 3

Ans: here, we have $f(t) = \begin{vmatrix} \cos t & t & 1 \\ 2 \sin t & t & 2t \\ \sin t & t & t \end{vmatrix}$

Expanding along R_1

$$\Rightarrow f(t) = \cos t(t^2 - 2t^2) - t(2t \sin t - 2t \sin t) + 1(2t \sin t - t \sin t)$$

$$\Rightarrow f(t) = -t^2 \cos t - 0 + t \sin t$$

$$\Rightarrow f(t) = -t^2 \cos t + t \sin t$$

Now, $\lim_{t \rightarrow 0} \frac{f(t)}{t^2} = \lim_{t \rightarrow 0} \left(\frac{-t^2 \cos t + t \sin t}{t^2} \right)$

$$\Rightarrow \lim_{t \rightarrow 0} \frac{f(t)}{t^2} = \lim_{t \rightarrow 0} \left(-\cos t + \frac{\sin t}{t} \right)$$

$$\Rightarrow \lim_{t \rightarrow 0} \frac{f(t)}{t^2} = -\lim_{t \rightarrow 0} \cos t + \lim_{t \rightarrow 0} \left(\frac{\sin t}{t} \right)$$

$$\Rightarrow \lim_{t \rightarrow 0} \frac{f(t)}{t^2} = -\lim_{t \rightarrow 0} \cos 0 + 1 \left[\lim_{t \rightarrow 0} \left(\frac{\sin t}{t} \right) = 1 \right]$$

$$\Rightarrow \lim_{t \rightarrow 0} \frac{f(t)}{t^2} = -1 + 1$$

$$\Rightarrow \lim_{t \rightarrow 0} \frac{f(t)}{t^2} = 0$$

Hence, option (A) is correct answer.

31. The maximum value of $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 + \sin \theta & 1 \\ 1 + \cos \theta & 1 & 1 \end{vmatrix}$ is ($\theta \in \mathbb{R}$).

(A) $\frac{1}{2}$

(B) $\frac{\sqrt{3}}{2}$

(C) $\sqrt{2}$

(D) $\frac{2\sqrt{3}}{4}$

Ans: here, we have $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 + \sin\theta & 1 \\ 1 + \cos\theta & 1 & 1 \end{vmatrix}$

Applying $C_1 \rightarrow C_1 - C_3$ & $C_2 \rightarrow C_2 - C_3$, we get

$$\Rightarrow \Delta = \begin{vmatrix} 0 & 0 & 1 \\ 0 & \sin\theta & 1 \\ \cos\theta & 0 & 1 \end{vmatrix}$$

Now, expanding along R_1

$$\Rightarrow \Delta = 1(0 - \sin\theta \cdot \cos\theta)$$

$$\Rightarrow \Delta = -\sin\theta \cdot \cos\theta$$

$$\Rightarrow \Delta = -\frac{2}{2}\sin\theta \cdot \cos\theta$$

$$\Rightarrow \Delta = -\frac{1}{2}\sin 2\theta$$

Also, we know that, $-1 \leq \sin 2\theta \leq 1$

$$\Rightarrow \frac{1}{2} \geq -\frac{1}{2}\sin 2\theta \geq -\frac{1}{2}$$

$$\Rightarrow -\frac{1}{2} \leq \Delta \leq \frac{1}{2}$$

Thus, maximum value of given determinant is $\frac{1}{2}$.

Hence, option (A) is correct answer.

32. If $f(x) = \begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix}$, then

(A) $f(a) = 0$

(B) $f(b) = 0$

(C) $f(0) = 0$

(D) $f(1) = 0$

Ans: here, we have $f(x) = \begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix}$

$$\Rightarrow f(a) = \begin{vmatrix} 0 & 0 & a-b \\ 2a & 0 & a-c \\ a+b & a+c & 0 \end{vmatrix} = 2a(a-b)(a+c) \neq 0$$

$$\Rightarrow f(b) = \begin{vmatrix} 0 & b-a & 0 \\ b+a & 0 & b-c \\ 2b & b+c & 0 \end{vmatrix} = -2b(b-a)(b-c) \neq 0$$

$$\Rightarrow f(0) = \begin{vmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{vmatrix} = abc - abc = 0$$

$$\Rightarrow f(1) = \begin{vmatrix} 0 & 1-a & 1-b \\ 1+a & 0 & 1-c \\ 1+b & 1+c & 0 \end{vmatrix} \neq 0$$

Hence, option (C) is correct answer.

33. If $A = \begin{bmatrix} 2 & \lambda & -3 \\ 0 & 2 & 5 \\ 1 & 1 & 3 \end{bmatrix}$, then A^{-1} exists if

(A) $\lambda = 2$

(B) $\lambda \neq 2$

(C) $\lambda \neq -2$

(D) None of these

Ans: here, we have $A = \begin{bmatrix} 2 & \lambda & -3 \\ 0 & 2 & 5 \\ 1 & 1 & 3 \end{bmatrix}$

Expanding along R_1

$$\Rightarrow |A| = 2(6 - 5) - \lambda(0 - 5) - 3(0 - 2)$$

$$\Rightarrow |A| = 2 + 5\lambda + 6$$

We know that, A^{-1} exists, if A is non- singular matrix i.e., $|A| \neq 0$

Therefore,

$$\Rightarrow |A| = 2 + 5\lambda + 6 \neq 0$$

$$\Rightarrow 5\lambda + 8 \neq 0$$

$$\Rightarrow 5\lambda \neq -8$$

$$\Rightarrow \lambda \neq -\frac{8}{5}$$

Hence, option (D) is correct answer.

34. If A and B are invertible matrices, then which of the following is not correct?

$$(A) \operatorname{adj} A = |A|.A^{-1}$$

$$(B) \det(A)^{-1} = [\det(A)]^{-1}$$

$$(C) (AB)^{-1} = B^{-1}A^{-1}$$

$$(D) (A+B)^{-1} = B^{-1} + A^{-1}$$

Ans: Since, A and B are invertible matrices. So, we can say that

$$(AB)^{-1} = B^{-1}A^{-1} \quad (\text{it is correct})$$

$$\text{Also, } A^{-1} = \frac{1}{|A|}(\operatorname{adj} A)$$

$$\Rightarrow \operatorname{adj} A = |A|.A^{-1} \quad (\text{it is correct})$$

$$\text{And } AA^{-1} = I$$

$$\Rightarrow |AA^{-1}| = |I|$$

$$\Rightarrow |A|.|A^{-1}| = 1$$

$$\Rightarrow |A^{-1}| = \frac{1}{|A|}$$

$$\Rightarrow \det(A)^{-1} = [\det(A)]^{-1} \quad (\text{it is correct})$$

$$\text{Now, } (A+B)^{-1} = B^{-1} + A^{-1}$$

$$\text{L.H.S} = (A+B)^{-1}$$

$$= \frac{1}{|A+B|}(\operatorname{adj} A + B)$$

$$\text{R.H.S} = B^{-1} + A^{-1}$$

$$= \frac{1}{|B|}(\operatorname{adj} B) + \frac{1}{|A|}(\operatorname{adj} A)$$

Thus, L.H.S $\neq$ R.H.S (it is incorrect)

Hence, option (D) is correct answer.

35. If x, y, z are all different from zero and $\begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1+z \end{vmatrix} = 0$, then value

$x^{-1} + y^{-1} + z^{-1}$ is

(A) xyz

(B) $x^{-1} \cdot y^{-1} \cdot z^{-1}$

(C) $-x - y - z$

(D) -1

Ans: here, we have $\begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1+z \end{vmatrix} = 0$

Applying $C_1 \rightarrow C_1 - C_3$ and $C_2 \rightarrow C_2 - C_3$

$$\Rightarrow \begin{vmatrix} x & 0 & 1 \\ -y & y & 1 \\ 0 & -z & 1+z \end{vmatrix} = 0$$

Expanding along R_1

$$\Rightarrow x(y + yz + z) + yz = 0$$

$$\Rightarrow xy + xyz + xz + yz = 0$$

$$\Rightarrow xy + xz + yz = -xyz$$

$$\Rightarrow \frac{xy + xz + yz}{xyz} = -1$$

$$\Rightarrow \frac{1}{z} + \frac{1}{y} + \frac{1}{x} = -1$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = -1$$

$$\Rightarrow x^{-1} + y^{-1} + z^{-1} = -1$$

Hence, option (D) is correct answer.

36. The value of the determinant $\begin{vmatrix} x & x+y & x+2y \\ x+2y & x & x+y \\ x+y & x+2y & x \end{vmatrix}$ is

(A) $9x^2(x+y)$

(B) $9y^2(x+y)$

(C) $3y^2(x+y)$

(D) $7x^2(x+y)$

Ans: here, we have $\begin{vmatrix} x & x+y & x+2y \\ x+2y & x & x+y \\ x+y & x+2y & x \end{vmatrix}$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$, we get

$$= \begin{vmatrix} 3(x+y) & x+y & x+2y \\ 3(x+y) & x & x+y \\ 3(x+y) & x+2y & x \end{vmatrix}$$

Taking common $3(x+y)$ from C_1

$$= 3(x+y) \begin{vmatrix} 1 & x+y & x+2y \\ 1 & x & x+y \\ 1 & x+2y & x \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$= 3(x+y) \begin{vmatrix} 1 & x+y & x+2y \\ 0 & -y & -y \\ 0 & y & -2y \end{vmatrix}$$

Expanding along C_1

$$= 3(x+y)(2y^2 + y^2)$$

$$= 3(x+y) \times 3y^2$$

$$= 9y^2(x+y)$$

Hence, option (B) is correct answer.

37. There are two values of a which makes determinant, $\Delta = \begin{vmatrix} 1 & -2 & 5 \\ 2 & a & -1 \\ 0 & 4 & 2a \end{vmatrix} = 86$, then

sum of these number is

(A) 4

(B) 5

(C) -4

(D) 9

Ans: here, we have $\Delta = \begin{vmatrix} 1 & -2 & 5 \\ 2 & a & -1 \\ 0 & 4 & 2a \end{vmatrix} = 86$

$$\Rightarrow \begin{vmatrix} 1 & -2 & 5 \\ 2 & a & -1 \\ 0 & 4 & 2a \end{vmatrix} = 86$$

Expanding along R_1

$$\Rightarrow 1(2a^2 + 4) + 2(4a - 0) + 5(8 - 0) = 86$$

$$\Rightarrow 2a^2 + 4 + 8a + 40 = 86$$

$$\Rightarrow 2a^2 + 8a + 44 = 86$$

$$\Rightarrow 2a^2 + 8a + 44 - 86 = 0$$

$$\Rightarrow 2a^2 + 8a - 42 = 0$$

$$\Rightarrow a^2 + 4a - 21 = 0$$

$$\Rightarrow a^2 + 7a - 3a - 21 = 0$$

$$\Rightarrow a(a + 7) - 3(a + 7) = 0$$

$$\Rightarrow (a + 7)(a - 3) = 0$$

$$\Rightarrow (a + 7) = 0 \text{ or } (a - 3) = 0$$

$$\Rightarrow a = -7 \text{ or } a = 3$$

Here, values of a are -7 and 3 .

Therefore, sum of these values = $-7 + 3 = -4$

Hence, option (C) is correct answer.

Fill in the blanks

1. If A is a matrix of order, 3×3 , then $|3A| = \dots\dots\dots$

Ans: here, A is a matrix of order, 3×3 .

And we know that, $|\lambda A| = \lambda^n |A|$, here n is the order of matrix and λ is a constant.

Therefore,

$$\Rightarrow |3A| = 3^3 |A| = 27|A|$$

Hence, If A is a matrix of order, 3×3 , then $|3A| = 27|A|$.

2. If A is invertible matrix of order 3×3 , then $|A^{-1}| = \dots\dots\dots$

Ans: here, A is invertible matrix of order 3×3

Therefore,

$$\Rightarrow AA^{-1} = I$$

$$\Rightarrow |AA^{-1}| = |I|$$

$$\Rightarrow |A| \cdot |A^{-1}| = 1$$

$$\Rightarrow |A^{-1}| = \frac{1}{|A|}$$

Hence, If A is invertible matrix of order 3×3 , then $|A^{-1}| = \frac{1}{|A|}$.

3. If $x, y, z \in \mathbb{R}$, then the value of determinant $\begin{vmatrix} (2^x + 2^{-x})^2 & (2^x - 2^{-x})^2 & 1 \\ (3^x + 3^{-x})^2 & (3^x - 3^{-x})^2 & 1 \\ (4^x + 4^{-x})^2 & (4^x - 4^{-x})^2 & 1 \end{vmatrix}$ is equal to

Ans: here, we have $\begin{vmatrix} (2^x + 2^{-x})^2 & (2^x - 2^{-x})^2 & 1 \\ (3^x + 3^{-x})^2 & (3^x - 3^{-x})^2 & 1 \\ (4^x + 4^{-x})^2 & (4^x - 4^{-x})^2 & 1 \end{vmatrix}$, where $x, y, z \in \mathbb{R}$,

Applying $C_1 \rightarrow C_1 - C_2$, we get

$$= \begin{vmatrix} (2^x + 2^{-x})^2 - (2^x - 2^{-x})^2 & (2^x - 2^{-x})^2 & 1 \\ (3^x + 3^{-x})^2 - (3^x - 3^{-x})^2 & (3^x - 3^{-x})^2 & 1 \\ (4^x + 4^{-x})^2 - (4^x - 4^{-x})^2 & (4^x - 4^{-x})^2 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 4 \cdot 2^x \cdot 2^{-x} & (2^x - 2^{-x})^2 & 1 \\ 4 \cdot 3^x \cdot 3^{-x} & (3^x - 3^{-x})^2 & 1 \\ 4 \cdot 4^x \cdot 4^{-x} & (4^x - 4^{-x})^2 & 1 \end{vmatrix} \quad [(a+b)^2 - (a-b)^2 = 4ab]$$

$$= \begin{vmatrix} 4 & (2^x - 2^{-x})^2 & 1 \\ 4 & (3^x - 3^{-x})^2 & 1 \\ 4 & (4^x - 4^{-x})^2 & 1 \end{vmatrix}$$

Taking common 4 from C_1

$$= 4 \begin{vmatrix} 1 & (2^x - 2^{-x})^2 & 1 \\ 1 & (3^x - 3^{-x})^2 & 1 \\ 1 & (4^x - 4^{-x})^2 & 1 \end{vmatrix} = 0$$

Since, determinant having all elements of any column or row gives value of determinant is zero.

Hence, then the value of determinant $\begin{vmatrix} (2^x + 2^{-x})^2 & (2^x - 2^{-x})^2 & 1 \\ (3^x + 3^{-x})^2 & (3^x - 3^{-x})^2 & 1 \\ (4^x + 4^{-x})^2 & (4^x - 4^{-x})^2 & 1 \end{vmatrix}$ is equal to 0.

4. If $\cos 2\theta = 0$, then $\begin{vmatrix} 0 & \cos \theta & \sin \theta \\ \cos \theta & \sin \theta & 0 \\ \sin \theta & 0 & \cos \theta \end{vmatrix}^2 = \dots\dots\dots$

Ans: here, we have $\cos 2\theta = 0$

$$\Rightarrow \cos 2\theta = \cos 90^\circ$$

$$\Rightarrow 2\theta = 90^\circ$$

$$\Rightarrow \theta = 45^\circ \dots\dots\dots (i)$$

Also, we have $\begin{vmatrix} 0 & \cos \theta & \sin \theta \\ \cos \theta & \sin \theta & 0 \\ \sin \theta & 0 & \cos \theta \end{vmatrix}^2$

Expanding along R_1

$$= [0 - \cos \theta (\cos^2 \theta - 0) + \sin \theta (0 - \sin^2 \theta)]^2$$

$$= [-\cos^3 \theta - \sin^3 \theta]^2$$

$$= [\cos^3 \theta + \sin^3 \theta]^2$$

$$= \left[\left(\frac{1}{\sqrt{2}} \right)^3 + \left(\frac{1}{\sqrt{2}} \right)^3 \right]^2$$

$$= \left[\frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \right]^2$$

$$= \left[\frac{1}{\sqrt{2}} \right]^2$$

$$= \frac{1}{2}$$

Hence, If $\cos 2\theta = 0$, then $\begin{vmatrix} 0 & \cos \theta & \sin \theta \\ \cos \theta & \sin \theta & 0 \\ \sin \theta & 0 & \cos \theta \end{vmatrix}^2 = \frac{1}{2}$.

5. If A is a matrix of order 3×3 , then $(A^2)^{-1} = \dots\dots\dots$

Ans: If A is a matrix of order 3×3 , then $(A^2)^{-1} = (A^{-1})^{-2}$.

6. If A is a matrix of order 3×3 , then number of minor in determinant of A are $\dots\dots\dots$

Ans: If A is a matrix of order 3×3 , then number of minor in determinant of A are 9. because, in a 3×3 matrix, there are 9 elements.

7. The sum of the products of elements of any row with the co-factors of corresponding elements is equal to

Ans: The sum of the products of elements of any row with the co-factors of corresponding elements is equal to value of determinant.

$$\text{Let } \Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Expanding along R_1

$$\Rightarrow \Delta = a_{11}(a_{22} \cdot a_{33} - a_{23} \cdot a_{32}) - a_{12}(a_{21} \cdot a_{33} - a_{23} \cdot a_{31}) + a_{13}(a_{21} \cdot a_{32} - a_{22} \cdot a_{31})$$

$$\Rightarrow \Delta = a_{11}(a_{22} \cdot a_{33} - a_{23} \cdot a_{32}) + a_{12}(a_{23} \cdot a_{31} - a_{21} \cdot a_{33}) + a_{13}(a_{21} \cdot a_{32} - a_{22} \cdot a_{31})$$

$$\Rightarrow \Delta = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

$\Rightarrow \Delta =$ sum of the products of elements of R_1 with their corresponding cofactors.

8. If $x = -9$ is a root of $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$, then other two roots are

Ans: here, we have $x = -9$ is a root of $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$

$$\Rightarrow \begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$$

Expanding along R_1 , we get

$$\Rightarrow x(x^2 - 12) - 3(2x - 14) + 7(12 - 7x) = 0$$

$$\Rightarrow x^3 - 12x - 6x + 42 + 84 - 49x = 0$$

$$\Rightarrow x^3 - 12x - 6x + 42 + 84 - 49x = 0$$

$$\Rightarrow x^3 - 67x + 126 = 0$$

$$\Rightarrow x^3 + 9x^2 - 9x^2 - 81x + 14x + 126 = 0$$

$$\Rightarrow x^2(x + 9) - 9x(x + 9) + 14(x + 9) = 0$$

$$\Rightarrow (x + 9)(x^2 - 9x + 14) = 0$$

$$\Rightarrow (x + 9)(x - 7)(x - 2) = 0$$

$$\Rightarrow x = -9, 7, 2$$

Hence, other two roots of $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$ are 7 and 2.

9. $\begin{vmatrix} 0 & xyz & x-z \\ y-x & 0 & y-z \\ z-x & z-y & 0 \end{vmatrix} = \dots\dots\dots$

Ans: here, we have $\begin{vmatrix} 0 & xyz & x-z \\ y-x & 0 & y-z \\ z-x & z-y & 0 \end{vmatrix}$

Applying $C_1 \rightarrow C_1 - C_3$, we get

$$= \begin{vmatrix} z-x & xyz & x-z \\ z-x & 0 & y-z \\ z-x & z-y & 0 \end{vmatrix}$$

Taking common $(z - x)$ from C_1

$$= (z - x) \begin{vmatrix} 1 & xyz & x - z \\ 1 & 0 & y - z \\ 1 & z - y & 0 \end{vmatrix}$$

Applying $[R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1]$, we get

$$= (z - x) \begin{vmatrix} 1 & xyz & x - z \\ 0 & -xyz & y - x \\ 0 & z - y - xyz & z - x \end{vmatrix}$$

Expanding along C_1

$$= (z - x) [-xyz(z - x) - (y - x)(z - y - xyz)]$$

$$= (z - x) [-xyz^2 + x^2yz - (yz - y^2 - xy^2z - xz + xy + x^2yz)]$$

$$= (z - x) [-xyz^2 - yz + y^2 + xy^2z + xz - xy]$$

$$= (z - x) [y^2 - yz + xy^2z - xyz^2 + xz - xy]$$

$$= (z - x) [y(y - z) + xyz(y - z) - x(y - z)]$$

$$= (z - x)(y - z)(y + xyz - x)$$

$$\text{Hence, } \begin{vmatrix} 0 & xyz & x - z \\ y - x & 0 & y - z \\ z - x & z - y & 0 \end{vmatrix} = (z - x)(y - z)(y + xyz - x).$$

$$10. \text{ If } f(x) = \begin{vmatrix} (1+x)^{17} & (1+x)^{19} & (1+x)^{23} \\ (1+x)^{23} & (1+x)^{29} & (1+x)^{34} \\ (1+x)^{41} & (1+x)^{43} & (1+x)^{47} \end{vmatrix} = A + Bx + Cx^2 + \dots, \text{ then } A = \dots\dots\dots$$

$$\text{Ans: here, we have } f(x) = \begin{vmatrix} (1+x)^{17} & (1+x)^{19} & (1+x)^{23} \\ (1+x)^{23} & (1+x)^{29} & (1+x)^{34} \\ (1+x)^{41} & (1+x)^{43} & (1+x)^{47} \end{vmatrix}$$

Taking common $(1+x)^{17}$, $(1+x)^{23}$, $(1+x)^{41}$ from R_1 , R_2 & R_3 respectively, we get

$$= (1+x)^{17} \cdot (1+x)^{23} \cdot (1+x)^{41} \begin{vmatrix} 1 & (1+x)^2 & (1+x)^6 \\ 1 & (1+x)^6 & (1+x)^{11} \\ 1 & (1+x)^2 & (1+x)^6 \end{vmatrix}$$

$$= 0.$$

Here, R_1 & R_3 are identical,

$$\text{Thus, } f(x) = \begin{vmatrix} (1+x)^{17} & (1+x)^{19} & (1+x)^{23} \\ (1+x)^{23} & (1+x)^{29} & (1+x)^{34} \\ (1+x)^{41} & (1+x)^{43} & (1+x)^{47} \end{vmatrix} = 0.$$

Hence, value of A is 0.

State True or False for the statements of the following Exercises:

11. $(A^3)^{-1} = (A^{-1})^3$, where A is square matrix and $|A| \neq 0$.

Ans: here, the given statement is true.

Since, $(A^n)^{-1} = (A^{-1})^n$, where $n \in \mathbb{N}$.

12. $(aA)^{-1} = \frac{1}{a} A^{-1}$, where a is any real number and A is a square matrix.

Ans: Since, we know that, if A is a non-singular square matrix, then for any scalar a (non-zero), aA is invertible such that

$$\Rightarrow (aA) = \left(\frac{1}{a} A^{-1} \right) = \left(a \cdot \frac{1}{a} \right) (AA^{-1}) \text{ i.e., } (aA) \text{ is inverse of } \left(\frac{1}{a} A^{-1} \right) \text{ or } (aA)^{-1} = \frac{1}{a} A^{-1},$$

where a is any non-zero scalar false.

In the above statement a is any real number. So, we can conclude that above statement is false.

13. $|A^{-1}| \neq |A|^{-1}$, where A is non – singular matrix.

Ans: Since, $|A^{-1}| = |A|^{-1}$, where A is non – singular matrix.

Hence, the given statement is false.

**14. If A and B are the matrices of order 3 and $|A|=5$, $|B|=3$, then $|2AB|$
 $27 \times 5 \times 3 = 405$.**

Ans: here, A and B are the matrices of order 3 and $|A|=5$, $|B|=3$

$$\Rightarrow |3AB| = 3^3 \times |A| \times |B|$$

$$\Rightarrow |3AB| = 27 \times 5 \times 3$$

$$\Rightarrow |3AB| = 405$$

Hence, given statement is true.

15.If the value of a third order determinant is 12, then the value of the determinants formed by replacing each element by its co – factor will be 144.

Ans: let A be the determinant, $|A|=12$

Also, we know that, if A is the square matrix of order n,

$$\text{then } |\text{Adj } A| = |A|^{n-1}$$

$$\Rightarrow |\text{Adj } A| = (12)^{3-1}$$

$$\Rightarrow |\text{Adj } A| = (12)^2$$

$$\Rightarrow |\text{Adj } A| = 144$$

Hence, the given statement is true.

16. $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix} = 0$, where a, b, c are in A.P.

Ans: here, we have $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$

Applying $C_1 \rightarrow C_1 - C_2$, we get

$$= \begin{vmatrix} -1 & x+2 & x+a \\ -1 & x+3 & x+b \\ -1 & x+4 & x+c \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - R_1$ & $R_3 \rightarrow R_3 - R_1$, we get

$$= \begin{vmatrix} -1 & x+2 & x+a \\ 0 & 1 & b-a \\ 0 & 2 & c-a \end{vmatrix}$$

Expanding along C_1

$$= -1[(c-a) - 2(b-a)]$$

$$= -1(c-a-2b+2a)$$

$$= -1(c-2b+a)$$

Since, a, b and c are in A.P.

Therefore, $2b = a + c$

$$= -1(2b-2b)$$

$$= 0.$$

Hence, given statement is true.

17. $|\text{adj. } A| = |A|^2$, where A is square matrix of order two.

Ans: here, A is square matrix of order two

We know that,

$$\Rightarrow A \cdot \text{Adj } A = |A| I$$

$$\Rightarrow |A \cdot \text{Adj } A| = ||A| I|$$

$$\Rightarrow |A| |\text{Adj } A| = |A|^n$$

$$\Rightarrow |\text{Adj } A| = \frac{|A|^n}{|A|}$$

$$\Rightarrow |\text{Adj } A| = |A|^{n-1}$$

Here, $n = 2$

$$\Rightarrow |\text{Adj } A| = |A|^{2-1}$$

$$\Rightarrow |\text{Adj } A| = |A|$$

Hence, the given statement is false.

18. The determinant $\begin{vmatrix} \sin A & \cos A & \sin A + \cos B \\ \sin B & \cos A & \sin B + \cos B \\ \sin C & \cos A & \sin C + \cos B \end{vmatrix}$ **is equal to zero.**

Ans: here, we have $\begin{vmatrix} \sin A & \cos A & \sin A + \cos B \\ \sin B & \cos A & \sin B + \cos B \\ \sin C & \cos A & \sin C + \cos B \end{vmatrix}$

$$= \begin{vmatrix} \sin A & \cos A & \sin A \\ \sin B & \cos A & \sin B \\ \sin C & \cos A & \sin C \end{vmatrix} + \begin{vmatrix} \sin A & \cos A & \cos B \\ \sin B & \cos A & \cos B \\ \sin C & \cos A & \cos B \end{vmatrix}$$

Since, the value of determinant having two identical rows and columns is zero. Therefore,

$$= 0 + \begin{vmatrix} \sin A & \cos A & \cos B \\ \sin B & \cos A & \cos B \\ \sin C & \cos A & \cos B \end{vmatrix}$$

$$= \begin{vmatrix} \sin A & \cos A & \cos B \\ \sin B & \cos A & \cos B \\ \sin C & \cos A & \cos B \end{vmatrix}$$

Taking common $\cos A$ and $\cos B$ from C_2 and C_3

$$= \cos A \cdot \cos B \begin{vmatrix} \sin A & 1 & 1 \\ \sin B & 1 & 1 \\ \sin C & 1 & 1 \end{vmatrix}$$

$$= 0$$

Since, the value of determinant having two identical rows and columns is zero.

Hence, given statement is true.

19. If the determinant $\begin{vmatrix} x+a & p+u & l+f \\ y+b & q+v & m+g \\ z+c & r+w & n+h \end{vmatrix}$ splits into exactly K determinants of order 3, each element of which contains only one term, then the value of K is 8.

Ans: here, we have $\begin{vmatrix} x+a & p+u & l+f \\ y+b & q+v & m+g \\ z+c & r+w & n+h \end{vmatrix}$

Now, splitting first row in the sum of two determinant

$$= \begin{vmatrix} x & p & l \\ y+b & q+v & m+g \\ z+c & r+w & n+h \end{vmatrix} + \begin{vmatrix} a & u & f \\ y+b & q+v & m+g \\ z+c & r+w & n+h \end{vmatrix}$$

Now, splitting second row in the sum of two determinants

$$= \begin{vmatrix} x & p & l \\ y & q & m \\ z+c & r+w & n+h \end{vmatrix} + \begin{vmatrix} x & p & l \\ b & v & g \\ z+c & r+w & n+h \end{vmatrix} + \begin{vmatrix} a & u & f \\ y & q & m \\ z+c & r+w & n+h \end{vmatrix} + \begin{vmatrix} a & u & f \\ b & v & g \\ z+c & r+w & n+h \end{vmatrix}$$

Similarly, we can split these 4 determinants in 8 determinants by splitting each one in two determinants.

Hence, the given statement is true.

20. Let $\Delta = \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} = 16$, then $\Delta_1 = \begin{vmatrix} p+x & a+x & a+p \\ q+y & b+y & b+q \\ r+z & c+z & c+r \end{vmatrix} = 32$.

Ans: here, we have $\Delta = \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix} = 16$ (i)

Now, $\Delta_1 = \begin{vmatrix} p+x & a+x & a+p \\ q+y & b+y & b+q \\ r+z & c+z & c+r \end{vmatrix}$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$, we get

$$\Rightarrow \Delta_1 = \begin{vmatrix} 2(p+x+a) & a+x & a+p \\ 2(q+y+b) & b+y & b+q \\ 2(r+z+c) & c+z & c+r \end{vmatrix}$$

Taking common 2 from C_1

$$\Rightarrow \Delta_1 = 2 \begin{vmatrix} (p+x+a) & a+x & a+p \\ (q+y+b) & b+y & b+q \\ (r+z+c) & c+z & c+r \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 - C_2$, and $C_2 \rightarrow C_2 - C_3$ we get

$$\Rightarrow \Delta_1 = 2 \begin{vmatrix} p & x-p & a+p \\ q & y-q & b+q \\ r & z-r & c+r \end{vmatrix}$$

Applying $C_2 \rightarrow C_1 + C_2$

$$\Rightarrow \Delta_1 = 2 \begin{vmatrix} p & x & a+p \\ q & y & b+q \\ r & z & c+r \end{vmatrix}$$

$$\Rightarrow \frac{\Delta_1}{2} = \begin{vmatrix} p & x & a \\ q & y & b \\ r & z & c \end{vmatrix} + \begin{vmatrix} p & x & p \\ q & y & q \\ r & z & r \end{vmatrix}$$

$$\Rightarrow \frac{\Delta_1}{2} = \begin{vmatrix} p & x & a \\ q & y & b \\ r & z & c \end{vmatrix} + 0$$

Since, the value of determinant having two identical rows and columns is zero.

$$\Rightarrow \Delta_1 = 2 \begin{vmatrix} p & x & a \\ q & y & b \\ r & z & c \end{vmatrix}$$

$$\Rightarrow \Delta_1 = 2 \cdot (-1)^2 \begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}$$

$$\Rightarrow \Delta_1 = 2 \times 16 = 32 \quad \text{[from eq (i)]}$$

Hence, the given statement is true.

20. The maximum value of $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1 & 1 & 1+\cos\theta \end{vmatrix}$ **is** $\frac{1}{2}$.

Ans: here, we have $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1 & 1 & 1+\cos\theta \end{vmatrix}$

Applying $C_1 \rightarrow C_1 - C_3$ & $C_2 \rightarrow C_2 - C_3$, we get

$$\Rightarrow \Delta = \begin{vmatrix} 0 & 0 & 1 \\ 0 & \sin\theta & 1 \\ -\cos\theta & -\cos\theta & 1 \end{vmatrix}$$

Now, expanding along R_1

$$\Rightarrow \Delta = 1(0 + \sin\theta \cdot \cos\theta)$$

$$\Rightarrow \Delta = \sin\theta \cdot \cos\theta$$

$$\Rightarrow \Delta = \frac{2}{2} \sin\theta \cdot \cos\theta$$

$$\Rightarrow \Delta = \frac{1}{2} \sin 2\theta$$

Also, we know that, $-1 \leq \sin 2\theta \leq 1$

$$\Rightarrow -\frac{1}{2} \leq \frac{1}{2} \sin 2\theta \leq \frac{1}{2}$$

$$\Rightarrow -\frac{1}{2} \leq \Delta \leq \frac{1}{2}$$

Thus, maximum value of given determinant is $\frac{1}{2}$.

Hence, given statement is true.